Multistrategy Multiobjective Evolutionary Algorithm-Based Unsupervised Band Selection for Hyperspectral Image Classification

Heming Jia, *Member, IEEE*, and Changsheng Wen

***Abstract*—Hyperspectral images (HSIs) contain abundant spectral information but also exhibit significant redundancy. Therefore, selecting bands that possess high information content and low interband correlation is essential. Evolutionary algorithms (EAs) have demonstrated strong capabilities in addressing band selection (BS) problems. However, many existing methods rely on single-objective optimization, which limits their ability to jointly optimize multiple characteristics of the selected band subsets. To address this limitation, this study proposes an improved multiobjective EA for BS (IMOEABS). The proposed algorithm employs interclass distance and information entropy (IE) as objective functions to identify bands with strong class discriminability and high information richness. It is built upon the dimension-wise special crowding distance (DSCD) nondominated sorting mechanism, which penalizes small index differences between adjacent bands to suppress redundancy in the decision space. Furthermore, the generation strategy of offspring solutions is guided by the population entropy, enhancing the exploration efficiency of the algorithm. Experimental results on three benchmark datasets demonstrate that IMOEABS achieves superior performance compared to state-of-the-art BS methods, when evaluated using support vector machine (SVM) and *K*-nearest neighbor (KNN) classifiers. These findings confirm the robustness and effectiveness of IMOEABS in hyperspectral BS tasks.**

***Index Terms*—Band selection (BS), dimension-wise special crowding distance (DSCD), evolutionary algorithms (EAs), hyperspectral imaging, multiobjective.**

I. Introduction

HYPERSPECTRAL imaging [1] is a powerful remote sensing technology that captures land surface information across dozens to hundreds of contiguous spectral bands. Each pixel in an hyperspectral images (HSI) contains not only spatial location information but also a continuous spectral reflectance curve, which reflects the physical and chemical properties of ground materials. This enables the accurate identification of subtle material variations, such as chlorophyll content in vegetation [2] or mineral absorption features [3]. Due to its ultrahigh spectral resolution, HSI can distinguish objects with highly similar spectral responses, which makes it an essential tool in precision agriculture [4], [5], mineral exploration [6], [7], environmental monitoring [8], [9], and medical imaging [10], [11], [12]. However, the advantage of HSI—its high spectral dimensionality—also introduces severe redundancy and computational burden. Redundant spectral bands not only increase data storage and processing costs but also deteriorate classification performance due to the well-known "curse of dimensionality" [13]. Therefore, dimensionality reduction (DR) [14] has become a fundamental preprocessing step to retain the most informative and discriminative spectral information while mitigating redundancy.

DR techniques can be broadly divided into feature extraction and feature selection. Feature extraction methods, such as principal component analysis (PCA) [15], linear discriminant analysis (LDA) [16], and nonlinear mappings like t-distributed stochastic neighbor embedding (t-SNE) [17], project high-dimensional data into a lower dimensional space. Although these methods effectively reveal intrinsic data structures, the transformed features are abstract and lack direct physical interpretability. In contrast, feature selection aims to identify a subset of the most informative original features [18], preserving the semantic and physical meaning of each band. In the hyperspectral domain, this process is known as band selection (BS), which reduces dimensionality while maintaining spectral interpretability and improving the efficiency of subsequent classification tasks.

Existing BS methods are mainly categorized into clustering-based [19], [20], ranking-based [21], and evolutionary algorithm (EA)-based methods [22]. Although clustering and ranking strategies are computationally efficient, they rely heavily on heuristic assumptions and often fail to balance information richness with redundancy suppression. EA-based approaches, in contrast, exhibit strong global optimization capability and are well suited to the nonconvex, multimodal nature of BS problems. Nevertheless, current EA-based BS frameworks still suffer from three major limitations. First, many treat bands as independent variables, neglecting interband correlations and thus producing redundant sub-

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Heming Jia is with the School of Information Engineering, Sanming University, Sanming 365004, China (e-mail: jiaheming@fjsmu.edu.cn).

Changsheng Wen is with the College of Information and Electrical Engineering, Heilongjiang Bayi Agricultural University, Daqing 163319, China (e-mail: wenchangsheng@byau.edu.cn).

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sets [23]. Second, conventional integer or binary encoding schemes are poorly aligned with the arithmetic behavior of evolutionary operators, easily causing boundary violations or repeated BSs during evolution. Third, traditional crowding-distance-based environmental selection mechanisms cannot adequately measure diversity in decision space, leading to premature convergence and poor coverage of the Pareto front (PF) [24]. These issues jointly constrain the balance between global exploration and local convergence, limiting both the quality and diversity of selected band subsets.

To address these challenges, this study proposes an improved multiobjective EA for BS (IMOEABS) that integrates encoding adaptation, enhanced diversity measurement, and entropy-driven operator selection into a unified optimization framework. The theoretical motivation behind IMOEABS lies in the intrinsic coupling between these components. The encoding mechanism defines the geometric structure of the search space and determines how effectively operators can explore it. The environmental selection metric regulates population diversity and suppresses redundant solutions. The adaptive operator selection mechanism dynamically balances exploration and exploitation according to the evolution state. Hence, integrating these elements is not an arbitrary combination but a necessary strategy to achieve efficient and stable optimization in hyperspectral BS.

Within this framework, a symmetric vertical spectral feature coding mechanism is designed to map band indices into a symmetric interval, aligning the search space with the arithmetic nature of evolutionary operations. This mitigates boundary violations and improves search stability. A dimension-wise special crowding distance (DSCD) is introduced to measure diversity across decision dimensions, enabling more accurate discrimination between individuals and effectively avoiding redundant band aggregation. Furthermore, an entropy-driven adaptive operator selection strategy is employed to automatically adjust mutation and crossover operators, thereby enhancing the balance between convergence speed and population diversity. To guide the optimization process, a dual-objective model is formulated by combining information entropy (IE) and the Bhattacharyya distance, ensuring that the selected bands exhibit both high information content and strong interclass separability.

By integrating these innovations, IMOEABS establishes a framework that addresses the fundamental deficiencies of existing BS methods, including redundancy persistence, weak interaction between encoding and operators, and insufficient diversity maintenance. Experimental validation demonstrates that this coordinated design substantially improves classification accuracy, convergence stability, and subset diversity across multiple hyperspectral datasets, providing a robust and generalizable solution for high-dimensional spectral data analysis.

The main contributions of this work are summarized as follows.

1) A novel multiobjective EA (MOEA) is developed to address the BS problem. This algorithm integrates quadratic interpolation (QI), differential evolution (DE), and an entropy-driven operator selection mechanism to enhance search performance and convergence efficiency.
2) A symmetric vertical spectral feature encoding mechanism is proposed. This encoding strategy is designed to align better with the characteristics of EA operators, thereby improving the effectiveness of the search process.
3) A DSCD metric is introduced, which evaluates crowding based on dimensional distances. This approach effectively suppresses the selection of redundant bands and enhances the diversity of candidate subsets.
4) A novel cost function is constructed by combining IE and the average interclass distance, which is computed using the Bhattacharyya distance. This dual-objective formulation ensures that selected bands exhibit both high information richness and strong class discrimination capability.

The structure of the article is organized as follows. Section II presents the related work, discussing and critically comparing existing hyperspectral BS methods. Section III introduces the DE algorithm and the multiobjective optimization problem (MOP). Section IV provides specific details about the IMOEABS method. Section V presents the experimental setup and general analysis. Section VI analyzes the performance of IMOEABS from various perspectives. Finally, Section VII offers conclusions.

II. Related Work

Existing BS methods for hyperspectral imaging can be broadly categorized into three methodological families: clustering-based, ranking-based, and EA-based approaches. Each category reflects a specific tradeoff between information preservation and redundancy reduction, and their respective limitations motivate the design of the proposed IMOEABS framework.

A. Clustering-Based Methods

Clustering-based BS methods group spectrally similar bands into clusters and then select representative bands from each cluster. For instance, HCBH [19] constructs a hierarchical structure of spectral bands through hyperbolic clustering, while NGNMF [25] refines grouped bands via the neighborhood band grouping (NBG) strategy and selects representative ones using normalized matched filtering. These approaches effectively reduce redundancy and are computationally efficient. However, they heavily depend on the clustering quality and are sensitive to noise, which may result in unstable or suboptimal grouping of correlated bands and degrade classification performance. Moreover, since clustering often focuses on spectral similarity, the discriminative capability between classes is rarely optimized explicitly.

B. Ranking-Based Methods

Ranking-based methods evaluate the importance of each band individually according to certain metrics and then select the top-ranked ones. For example, MPGAE [21] measures

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the spatial gradient magnitude of each band to capture local information variation and suppress redundancy through thresholding. MVPCA [23], a variance-based method derived from PCA, prioritizes bands contributing most to the data variance. Such approaches are simple, interpretable, and computationally efficient. Nevertheless, they often neglect the interband dependency, meaning that multiple highly correlated bands may be simultaneously selected, leading to redundant subsets and reduced classification diversity. Furthermore, most ranking-based schemes cannot adaptively balance the tradeoff between information richness and redundancy during selection.

C. EA-Based Methods

EA-based BS methods formulate the problem as an optimization task and employ population-based search to explore diverse subsets. Wang et al. [26] proposed a Tri-evolutionary mechanism with information interaction of DE (TEDE), where two independent populations evolve separately and later interact to exchange information. Wu et al. [27] developed an unsupervised method based on heterogeneous Cuckoo search and matched filter (HCS-MF), which integrates a quantum mechanism to enhance late-iteration search and adopts an NBG-based mapping strategy to select high-SNR and low-similarity bands.

Other representative single-objective EA methods include HGWO [28], IROABS [29], SSATA [30], and SSGA [31], which optimize criteria such as band variance [32], interclass distance [33], or intraclass distance [34]. Although effective, single-objective EAs typically emphasize one optimization metric, making them less suitable for the inherently multiobjective nature of BS.

To address this limitation, MOEAs have been widely adopted. Yong et al.'s MABC-BS [22] employ multidirectional search and adaptive mutation to balance spectral information and interband correlation. Ou et al. [24] proposed MOCS-BS, which enhances Cuckoo Search with population crowding and group-based crossover to improve diversity. Wang et al. [35] introduced MEMT-HBS, using a multitasking framework to optimize uniform band distribution and class separability across different subset sizes via intertask knowledge transfer. Zhang et al.'s APSO-BS [36] combine entropy and structural similarity within an adaptive PSO to dynamically adjust search strategies and repair redundant subsets. Similar multiobjective methods, including MACOBS [37], MOBSTD [38], BOMBS [39], MOMVOBS [40], and MOSCA [41], further improve convergence and diversity through hybrid objective formulations and adaptive operators.

Despite these advances, several challenges persist among existing EA-based BS methods. First, the coupling between encoding and search operators remains weak, reducing exploration efficiency and hindering population diversity. Second, most environmental selection mechanisms rely solely on objective-space crowding measures, failing to reflect decision-space diversity such as interband spacing. Third, redundancy suppression is often implemented as postprocessing rather than integrated into the evolutionary process, which limits robustness and generalization. These limitations motivate the design of the IMOEABS framework, which introduces symmetric encoding, a DSCD, and entropy-driven operator selection to achieve a better balance between information preservation and redundancy reduction.

D. Summary and Motivation

In summary, existing hyperspectral BS methods exhibit distinct strengths and limitations. Clustering-based approaches efficiently reduce redundancy by grouping spectrally similar bands, but they heavily rely on clustering quality and often fail to explicitly optimize class discrimination. Ranking-based methods are simple, interpretable, and computationally efficient, yet they generally ignore interband dependencies, which can lead to redundant subsets and insufficient balancing of information richness and redundancy. EA-based methods, especially multiobjective variants, provide flexible optimization frameworks and consider multiple conflicting objectives, but challenges such as weak coupling between encoding and search operators, insufficient environmental selection, and incomplete redundancy suppression persist.

These observations indicate that current methods are either limited in handling high redundancy, lack effective integration of information preservation and discrimination, or fail to fully exploit the multiobjective potential of evolutionary search. Therefore, it is necessary to develop an improved BS framework that can simultaneously preserve informative spectral content, reduce redundancy, and maintain diversity within selected subsets. Motivated by these limitations, this article proposes the IMOEABS framework, a multiobjective evolutionary BS approach that explicitly integrates encoding design, diversity preservation, and adaptive operator control into a unified optimization process. Specifically, IMOEABS first represents candidate band subsets using a symmetric vertical spectral feature coding scheme, which reshapes the search space into a symmetric interval and improves the stability of evolutionary operations. Then, during environmental selection, a DSCD is employed to evaluate diversity directly in the decision space by penalizing small index differences between selected bands, thereby suppressing redundancy among adjacent spectral bands. Finally, an entropy-driven operator selection mechanism dynamically guides the offspring generation process by switching among different evolutionary operators according to the population entropy, enabling an adaptive balance between global exploration and local exploitation throughout the optimization.

III. Basic Knowledge

A. DE Operator

DE [42] shares a structural similarity with genetic algorithms (GAs) [31] and represents a simple yet effective global optimization technique. Inspired by the process of biological evolution, DE achieves optimization through three core operations: mutation, crossover, and selection. The mutation operator generates new candidate solutions by utilizing the difference vectors between randomly selected individuals within the population, thereby introducing directional perturbations.

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The crossover operator enhances population diversity by mixing components of the mutated vector with those of the target vector. Finally, the selection operator adopts a greedy strategy to retain the better-performing individual, as determined by the fitness function, for inclusion in the next generation. The variation operation mode of DE/current to best/1 [43] as follows:

$$V_i = X_i + F \cdot (X_{\text{best}} - X_i) + F \cdot (X_{r1} - X_{r2}) \tag{1}$$

where V_i represents the generated mutation vector; X denotes the entire population; X_{best} is the optimal solution; X_{r1} and X_{r2} are two randomly selected distinct individuals; r_1, r_2, and i are mutually distinct indices; and F is the scaling factor.

The formula of the crossover operation is as follows:

$$U_{i,j} = \begin{cases} V_{i,j}, & \text{if rand}\,(j) \le \text{CRor } j = j_{\text{rand}} \\ X_{i,j}, & \text{otherwise} \end{cases} \tag{2}$$

where U_i denotes the individual after mutation, rand(j) is a random number uniformly distributed between 0 and 1, CR is the crossover factor indicating the crossover rate, and j_{rand} is a randomly selected integer from the set $\{1, 2, 3, \ldots, \text{dim}\}$.

The formula for the selection operation is as follows:

$$\text{Off}_i = \begin{cases} U_i^t, & \text{if } f(U_i) \le f(X_i) \\ X_i^t, & \text{otherwise.} \end{cases} \tag{3}$$

B. Multiobjective Optimization Problems

MOPs involve simultaneously optimizing multiple conflicting objectives. In this article, the MOP is formulated as a maximization problem, defined by the following equation:

$$\text{maximize Fit}\,(x) = (f_1(x), \ldots, f_m(x)) \tag{4}$$

where x represents a feasible solution in the solution space, and f denotes the objective vector in the objective space. In general, it is not possible to obtain a solution that simultaneously maximizes all objectives. Therefore, the core challenge in multiobjective optimization is to obtain a set of well-balanced solutions that achieve a tradeoff among the objectives in the objective vector.

Fast nondominated sorting [44] is a key technique in the field of multiobjective optimization, primarily used to efficiently rank individuals in a population based on Pareto dominance. Given two solutions x_1 and x_2 with corresponding objective vectors Fit_1 and Fit_2, if every objective in Fit_1 is better than the corresponding one in Fit_2, then Fit_1 is said to dominate Fit_2, and Fit_2 is dominated by Fit_1. If only some objectives in Fit_1 are better than those in Fit_2, then there is no dominance relationship between Fit_1 and Fit_2. The set of solutions in the search space that are not dominated by any other solutions forms the PF.

IV. Proposed IMOEABS Method

This section proposes a method, denoted as MOEABS, for BS. The method incorporates two cost functions, which are designed based on the interclass relationships of band information and IE. In addition, an MOEA is employed specifically to solve the BS problem.

A. Objective Function

To address the BS problem using EAs, it is essential to design an effective evaluation function. This function should quantify the interclass distance and information redundancy among bands in the HSI, thereby enhancing classification performance.

Interclass distance is a crucial metric in hyperspectral image analysis, as it reflects the separability between different classes for a given band. A larger interclass distance indicates that the spectral characteristics of different classes are more distinct, implying that the band is more informative for classification tasks. Motivated by the Bhattacharyya distance, this article proposes an interclass distance calculation method based on the Bhattacharyya distance. The mean Bhattacharyya distance (MBD) is defined as the first objective function, denoted as f_1

$$f_1 = \frac{1}{K} \sum_{m=1}^{K} \sqrt{3\left(1 - e^{-\text{BD}_m/2}\right)} \tag{5}$$

where K denotes the total number of bands considered for evaluation, and BD_m represents the cumulative Bhattacharyya distance for the mth band. It is computed by summing the distances between all possible pairs of classes. For each class pair (i, j), the Bhattacharyya distance accounts for both the difference in their mean spectral values and the variance of each class, capturing the degree of overlap between their distributions. Formally, it is calculated as

$$\text{BD}_m = \sum_{i=1}^{C} \sum_{j=i+1}^{C} \left(\frac{(\mu_i - \mu_j)^2}{4\left(\sigma_i^2 + \sigma_j^2 + \text{eps}\right)} + \frac{1}{2} \ln \left(\frac{\sigma_i^2 + \sigma_j^2 + \text{eps}}{2\sqrt{\sigma_i^2 + \sigma_j^2 + \text{eps}}} \right) \right) \tag{6}$$

where C is the number of classes, and μ_i and μ_j are the mean values of classes i and j, respectively. And σ_i^2 and σ_j^2 are the variances of classes i and j, respectively.

The Bhattacharyya distance measures class separability by quantifying the degree of overlap between two probability distributions. It effectively captures both the mean difference and the covariance structure of spectral data. In contrast to Euclidean distance, which only considers the difference in means, the Bhattacharyya distance [45] provides a more comprehensive representation of the distribution shape and overlap. A larger BD value indicates less overlap between the spectral distributions of two classes, stronger class separability, and greater contribution to classification. For example, in hyperspectral BS, bands with high BD values can significantly enhance classification accuracy, whereas bands with low BD values, which exhibit high spectral overlap and weak discrimination, can be discarded.

IE [46] reflects the richness of information in an HSI band. Generally, a higher IE indicates that the band carries more information and has lower redundancy. The formula

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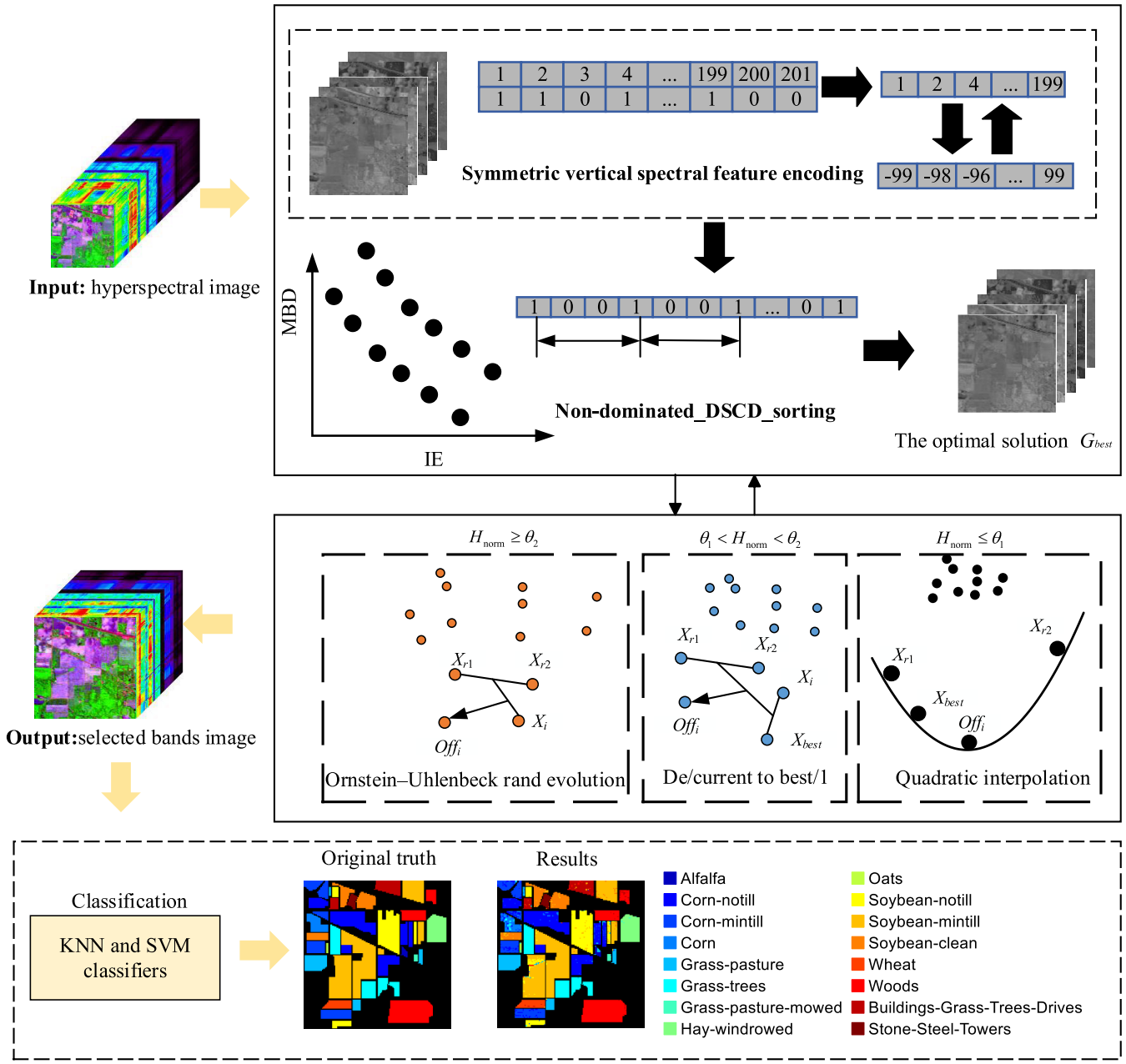


Fig. 1. Framework of IMOEABS method.

for the second objective function, IE f_2, is given as follows:

$$f_2 = -\frac{1}{K}\sum_{m=1}^{K} p(x_m)\log p(x_m) \tag{7}$$

$$p(x_m) = \sum_{l=0}^{255}\frac{\text{sum}(l)}{h \times w} \tag{8}$$

where K denotes the total number of bands considered for evaluation. h and w represent the height and width of the hyperspectral image, respectively, and thus $h \times w$ gives the total number of pixels in the band x_m. The term sum(l) counts the number of pixels with grayscale value l in band x_m, so that $p(x_m)$ corresponds to the probability of each grayscale value within the band. This definition allows the calculation of the entropy-based objective function f_2, which measures the information content of each band.

B. Framework of IMOEABS Method

This section provides a comprehensive description of the proposed IMOEABS method. It emphasizes the enhancements introduced to the EA and explains how the improved multiobjective DE algorithm (IMOEA) is employed to address the BS problem. The overall structure of the IMOEABS framework is illustrated in Fig. 1. After initializing the population, each solution is mapped back to an integer-coded representation through the symmetric vertical spectral feature encoding mechanism. According to the population's IE, different update strategies are applied to enhance population diversity. The updated and original populations are then merged and ranked

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Algorithm 1 IMOEABS

Input:
N: population size
Max_EFs: maximum iterations number
I: HSI data
Output:
G_{best}: a selected band subset

```
Initialize population by (9)
Encode candidate solutions (10)
while EFs < Max_EFs do
    Calculate H_norm by (21)
    Calculate θ1 and θ2 by (24)
    if H_norm ≥ θ2 then
        Generate mutation vector V by (18)
        Off ← Generate offspring by (2) and (3)
    else if θ1 < H_norm < θ2 then
        Generate mutation vector V by DE/current-to-best/1
        Off ← Generate offspring by (2) and (3)
    else
        Off ← Generate offspring by (17)
    end if
    Fit ← evaluate Off by (5) and (7)
    Sorted_Pop ← Non-dominated-DSCD-sorting(Off ∪ X, Fit) by Algorithm 2
    X ← top N candidate solutions from sorted_pop
end while
G_best ← top 1 candidate solution from X
G_best ← decoding the candidate solution by (11)
return G_best
```

using the nondominated DSCD sorting strategy to generate offspring. Once the maximum number of evaluations is reached, the final population is sorted again using the same criterion, and the optimal band subset is selected for HSI classification.

The pseudocode of the IMOEABS algorithm is presented in Algorithm 1. Together, Fig. 1 and Algorithm 1 provide a detailed overview of the implementation and operational process of the proposed method.

1) Coding Method: An HSI is represented as a $h \times w \times b$ dataset, where b denotes the number of spectral bands, and $h \times w$ represents the spatial dimensions of each band. Li et al. [47] introduce a vertical spectral feature coding mechanism based on band index values. Suppose the goal is to select dim bands, where each solution is represented by a vector x_i, consisting of band indices: $x_i = [x_1, x_2, \ldots, x_J, \ldots, x_{\text{dim}}]$. Each x_j is an integer between 1 and b, corresponding to the index of a selected band. The initialization of the solution is given by the following equation:

$$X = \text{floor}\left((\text{rand}\,(N, \text{dim}) \times (ub - lb) + lb)\right) \tag{9}$$

where N denotes the population size. The lower and upper bounds are represented by lb and ub, respectively, where lb is typically set to 1, and ub corresponds to the total number of bands in the HSI.

Compared with traditional binary coding, the vertical spectral feature coding mechanism reduces the dimensionality of the problem. In addition, it fixes the number of selected bands, thereby improving the computational efficiency of the algorithm. However, due to the asymmetry between the upper and lower bounds, EAs are prone to boundary violations during the evolutionary process.

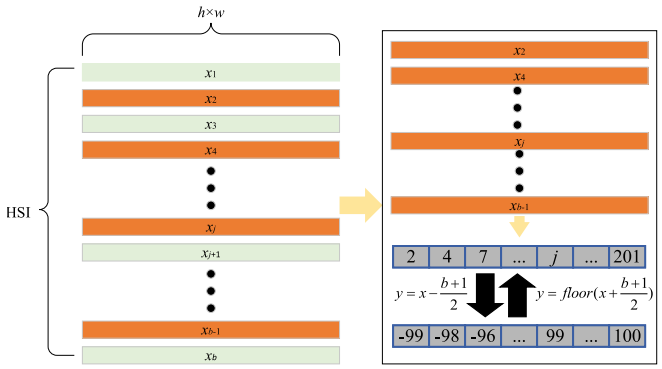


Fig. 2. Code method.

For example, let X_1 = [2, 5, 22, 33, 44] and X_2 = [3, 10, 28, 40, 60]. The result of $X_2 - X_1$ is $[1, 5, 6, 7, 16]$, which remains within bounds. In contrast, X_1 - X_2 yields $[-1, -5, -6, -7, -16]$, clearly violating the lower bound. If all elements of the resulting solution are clipped back to the lower bound, the result becomes $[1, 1, 1, 1, 1]$. It means the same band is repeatedly selected—rendering the solution ineffective.

To mitigate this issue, this article proposes a symmetric vertical spectral feature coding mechanism based on symmetrical interval conversion. Specifically, each solution is mapped from the original index interval $[1, b]$ to a symmetrical interval of $[-(b-1)/2, (b-1)/2]$ as defined in (10). The encoding process of this symmetrical interval conversion is illustrated in Fig. 2. In Fig. 2, the orange bars represent the selected bands, which are encoded according to their index values. Assuming that the dataset contains 201 spectral bands, the lower and upper bounds of each dimension are set to 1 and 201, respectively. Each dimension is then mapped to the range $[-100, 100]$ to reduce the frequency of boundary retraction during the evolutionary process, thereby improving search stability. After the evolutionary operations are completed, the results are remapped back to the original interval $[1, b]$ using (11) to obtain the final selected band indices

$$y = x - \frac{b+1}{2} \tag{10}$$

$$y = \text{floor}\left(x + \frac{b+1}{2}\right). \tag{11}$$

2) Dimension-Wise Special Crowding Distance: Wang et al. [48] noted that adjacent spectral bands often exhibit high similarity. To address this issue and prevent the selection of adjacent bands, a DSCD is proposed, building upon the traditional SCD.

SCD [49] is a type of crowding distance originally developed for multimodal optimization problems to maintain diversity not only in the objective space but also in the decision space. DSCD adapts this concept for the multiobjective BS task by computing the crowding distance among decision variables—specifically, the band indices.

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Algorithm 2 Nondominated-DSCD-Sorting

Input:
Pop: population
Fit: Object values of population
Output:
$Sorted_Pop$: a sorted population
non-dominated levels Fit_1, Fit_2, ..., Fit_{num} ← non-dominated sorting on Pop (num is the number of levels)
while $j < num$ **do**
  Calculate DSCD by (16)
end while
$Sorted_Pop$ ← Sort the population Pop based on their nondominated rankings and DSCD values
return: $Sorted_Pop$

In DSCD, the crowding distance between adjacent bands is penalized, while combinations involving bands with larger index gaps are rewarded with higher distance values. This encourages the selection of more diverse bands and reduces redundancy. Consequently, band combinations with greater index intervals are favored during the selection process, enhancing the diversity of the decision space in multiobjective optimization.

The DSCD calculation formula is defined as (16)

$$\mathrm{CD}_{i,x} = \sqrt{\frac{1}{N}\sum_{j=1}^{\dim}\left(X_{i,j}-\mu_i\right)^2} \tag{12}$$

$$\mathrm{CD}_{i,f} = \frac{\left|F_{i-1,1}-F_{i+1,1}\right|}{\mathrm{Max}\left(F_1\right)-\mathrm{Min}\left(F_1\right)} + \frac{\left|F_{i-1,2}-F_{i+1,2}\right|}{\mathrm{Max}\left(F_2\right)-\mathrm{Min}\left(F_2\right)} \tag{13}$$

$$\mathrm{CD}_{i,x} = \frac{\mathrm{CD}_{i,x}-\mathrm{Min}\left(\mathrm{CD}_x\right)}{\mathrm{Max}\left(\mathrm{CD}_x\right)-\mathrm{Min}\left(\mathrm{CD}_x\right)} \tag{14}$$

$$\mathrm{CD}_{i,f} = \frac{\mathrm{CD}_{i,f}-\mathrm{Min}\left(\mathrm{CD}_f\right)}{\mathrm{Max}\left(\mathrm{CD}_f\right)-\mathrm{Min}\left(\mathrm{CD}_f\right)} \tag{15}$$

$$\mathrm{DSCD}_i = \begin{cases} \mathrm{Max}\left(\mathrm{CD}_{i,x},\mathrm{CD}_{i,f}\right), & \text{if } \mathrm{CD}_{i,x} > \mathrm{CD}_{\mathrm{avg},x} \\ & \text{or} \\ & \mathrm{CD}_{i,f} > \mathrm{CD}_{\mathrm{avg},f} \\ \mathrm{Min}\left(\mathrm{CD}_{i,x},\mathrm{CD}_{i,f}\right), & \text{otherwise} \end{cases} \tag{16}$$

where $\mathrm{CD}_{i,x}$ denotes the dispersion of decision variables in solution x_i, and let μ_i be the average value of x_i. The term $\mathrm{CD}_{i,f}$ denotes the crowding distance of the objective vector corresponding to solution x_i.

Furthermore, $\mathrm{CD}_{\mathrm{avg},x}$ and $\mathrm{CD}_{\mathrm{avg},f}$ are the mean values of the dispersion of decision variables and the crowding distances in the objective space, respectively, across the entire population.

When both $\mathrm{CD}_{i,x}$ and $\mathrm{CD}_{i,f}$ are smaller than their corresponding average values, it indicates that solution x_i exhibits a small index difference and low crowding distance in the objective space. Such solutions contribute little to diversity and are considered redundant; therefore, they should be eliminated during selection.

3) Quadratic Interpolation: QI [50] is a numerical technique used to approximate the value of a function within an interval defined by three known data points. This method constructs a quadratic polynomial (such as a second-degree polynomial) that passes through the three selected points and uses it to estimate the function value at an intermediate location.

In the context of this work, two solutions x_{r1} and x_{r2} are randomly selected, along with the best solution p_{best}, which is obtained based on nondominated sorting using the DSCD. These three solutions are used to fit a quadratic polynomial of the form $p(x) = ax^2 + bx + c$. The extremum of this polynomial is then computed and taken as a new candidate solution for the evolutionary process. As shown in the following equation:

$$\mathrm{Off}_i = \frac{1}{2}\cdot\frac{\begin{array}{l} f\left(X_{r1}\right)\left(X_2^2-P_{\mathrm{best}}^2\right)+f\left(X_2\right)\left(P_{\mathrm{best}}^2-X_{r1}^2\right) \\ +f\left(P_{\mathrm{best}}\right)\left(X_{r1}^2-X_2^2\right) \end{array}}{\begin{array}{l} f\left(X_{r1}\right)\left(X_2-P_{\mathrm{best}}\right)+f\left(X_2\right)\left(P_{\mathrm{best}}-X_{r1}\right) \\ +f\left(P_{\mathrm{best}}\right)\left(X_{r1}-X_2\right) \end{array}} \tag{17}$$

where f denotes the sum of the two objective values, f_1 and f_2, both of which have been normalized.

4) Ornstein–Uhlenbeck Rand Evolution, OURE: The OURE operator is designed based on the Ornstein–Uhlenbeck stochastic process [51]. By incorporating the mean-reverting drift and random noise characteristics of the OU process into the mutation operation, the operator enhances the local convergence and stability of the algorithm while preserving the global search capability of DE. The formulation of the OU-based mutation strategy is (18)

$$\begin{aligned} V_{i,j} &= X_{i,j}e^{-\theta} + u_i\left(1-e^{-\theta}\right) \\ &\quad + F\left(X_{r1j}-X_{r2j}\right) + \sigma\sqrt{\frac{1-e^{-2\theta}}{2\theta}}\xi \end{aligned} \tag{18}$$

$$\theta = 0.1 + 0.9\times\left(\frac{t}{T}\right)^2 \tag{19}$$

$$\sigma = 0.1 + 0.4\times\exp\left(-\lambda\times\frac{t}{T}\right). \tag{20}$$

In the expression, the term $x_i^t e^{-\theta} + u_i(1-e^{-\theta})$ represents the mean-reversion component, which guides the solution toward the optimal region and enhances convergence efficiency. The term $\sigma((1-e^{-2\theta})/(2\theta))^{1/2}\xi$ represents the stochastic noise, introducing random jumps with finite variance to prevent premature convergence to local optima. Here, ξ is a normally distributed random variable. The parameters θ and σ are functions of the iteration index t, designed to adaptively balance exploration and exploitation: during the early stages, the algorithm maintains weak mean reversion and strong noise to enhance global exploration; in the later stages, it transitions to strong mean reversion and weak noise to accelerate local convergence.

5) Entropy-Driven Operator Selection Mechanism: The operator selection mechanism based on population entropy dynamically guides the switching and parameter control of evolutionary operators by quantifying population diversity. This mechanism aims to maintain a balance between global exploration and local exploitation. IE serves as a measure of the "disorder" or diversity within the population: high entropy indicates greater diversity and suggests the need for exploration, while low entropy reflects limited diversity and

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implies a focus on exploitation. The population entropy is calculated using (21)

$$H_{\text{norm}} = \frac{H - H_{\min}}{H_{\max} - H_{\min} + \text{eps}} \tag{21}$$

$$H = \frac{1}{\dim} \sum_{j=1}^{\dim} \left(- \sum_{i=1}^{|\text{uval}s_j|} \frac{n_{ij}}{N} \log_2 \left(\frac{n_{ij}}{N} + \text{eps} \right) \right) \tag{22}$$

where $H_{\max} = \log_2 ub$ represents the maximum entropy, and $H_{\min} = 0$ denotes the minimum entropy. The term $|\text{uval}_{s_j}|$ indicates the number of unique values in the population along the jth dimension, and n_{ij} denotes the frequency of each distinct value in that dimension. Based on the calculated entropy values, the population dynamically selects different location update operators. The specific selection strategy is described as (23)

$$\begin{cases} \text{If } H_{\text{norm}} \le \theta_1, & \text{Use QI} \\ \text{If } H_{\text{norm}} \ge \theta_2, & \text{Use OURE} \\ \text{Otherwise}, & \text{Use DE/current-to-best/1/bin} \end{cases} \tag{23}$$

$$\theta_1 = \max\left(0, \mu_H - \sigma_H \exp\left(-|\Delta H|\right)\right)$$

$$\theta_2 = \min\left(1, \mu_H + \sigma_H \left(1 - \exp\left(-|\Delta H|\right)\right)\right) \tag{24}$$

where thresholds θ_1 and θ_2 are two adaptive parameters determined based on the mean μ_H and variance σ_H of historical entropy values. The term ΔH represents the rate of change between the current population entropy and that of the previous generation.

C. Complexity Analysis of IMOEABS

The computational complexity of the IMOEABS method primarily arises from population updating and nondominated DSCD sorting. The complexity of population updating is $O(N \times \dim \times \text{Max_EFs})$. The complexity of nondominated DSCD sorting includes two components: nondominated sorting with complexity $O(2 \times N^2 \times \text{Max_EFs})$, and the calculation of DSCD distances with complexity $O(N \times \log N \times \text{Max_EFs} + N \times \dim \times \text{Max_EFs})$. In summary, the overall computational complexity of the IMOEABS method is approximately $O(N^2 \times \text{Max_EFs})$.

V. Experiments and Results

A. Datasets Description

This section evaluates the performance of the IMOEABS method using three benchmark hyperspectral datasets: the Salinas dataset, the Kennedy Space Center dataset, and the Botswana dataset.

1) Salinas Dataset: The Salinas dataset is a widely used benchmark in hyperspectral remote sensing. It was acquired by the AVIRIS sensor and consists of 224 spectral bands, with a spatial resolution of 3.7 m and an image size of 512 × 217 pixels. The spectral range spans from 400 to 2500 nm. This dataset covers various land cover types, corresponding to 16 ground-truth classes. To improve data quality, 20 water absorption bands are typically removed during preprocessing. Fig. 3 illustrates the 3-D visualization and the ground-truth map of the Salinas dataset.

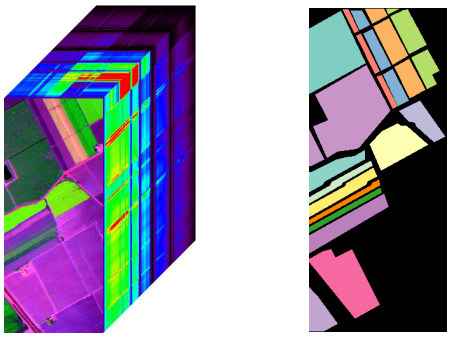


Fig. 3. Hyperspectral image cube and ground truth of Salinas. (a) Hyperspectral image cube. (b) Ground truth.

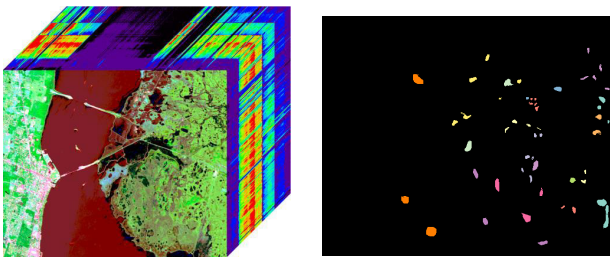


Fig. 4. Hyperspectral image cube and ground truth of Kennedy Space Center. (a) Hyperspectral image cube. (b) Ground truth.

2) Kennedy Space Center Dataset: The Kennedy Space Center dataset contains hyperspectral imagery collected by the AVIRIS sensor over the Kennedy Space Center in Florida, USA. It comprises 224 spectral bands with an image size of 512 × 614 pixels, and covers 13 land cover classes. The spectral range extends from 400 to 2500 nm. Due to significant water absorption effects in some bands, 58 bands are typically removed during preprocessing, resulting in 176 bands used for experiments. Fig. 4 shows the 3-D visualization and ground-truth map of the KSC dataset.

3) Botswana Dataset: The Botswana dataset is a hyperspectral dataset acquired by the Hyperion sensor aboard NASA's EO-1 satellite over the Okavango Delta, Botswana, on May 31, 2001. It consists of 242 spectral bands, each with an image size of 1476 × 256 pixels, and covers a wavelength range from 400 to 2500 nm. The dataset includes 14 distinct land cover types. Due to water absorption and sensor noise in certain bands—specifically bands 10–55 and 82–97—these are typically excluded during preprocessing, leaving 145 bands used in the experiments. Fig. 5 illustrates the 3-D visualization and ground-truth map of the Botswana dataset.

B. Experiment Description

To evaluate the practical performance of the proposed IMOEABS method, we compare it against seven representative BS algorithms from various categories: one EA-based method (IROABS) [29], one ranking-based method (MVPCA) [23], one clustering-based method (EFDPC) [20], and four hybrid clustering–ranking methods (NGNMF [25],

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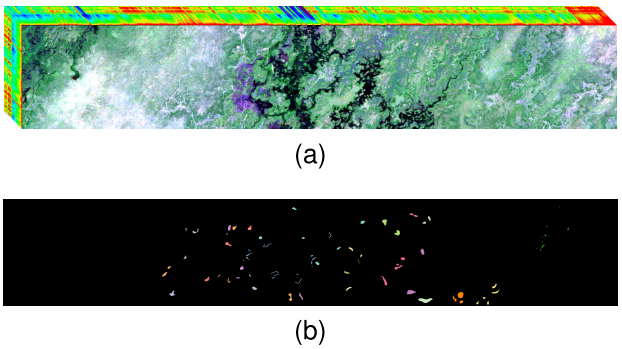


Fig. 5. Hyperspectral image cube and ground truth of Botswana. (a) Hyperspectral image cube. (b) Ground truth.

FNGBS [48], SPMCBS [52], ASPSMN [53], and SLFLRSC [54]). The details of these methods are as follows.

1) *IROABS:* An improved ROA-based BS method that defines a cost function based on class separability and IE. The ROA algorithm is enhanced via hybrid evolutionary operators, multistage mutation mechanisms, and guided learning. The final band subset is obtained by optimizing the proposed cost function.
2) *EFDPC:* This method enhances the FDPC clustering algorithm by weighting normalized local density and intracluster distance for each band. The truncation threshold is adaptively adjusted using exponential learning, and an outlier detection criterion is used to determine the optimal number of bands.
3) *NGNMF:* Combines clustering and ranking by grouping highly correlated bands and applying a normalized matched filter ranking strategy within each group, effectively reducing redundancy and enhancing classification performance.
4) *FNGBS:* Employs a coarse-to-fine grouping strategy to mine spectral contextual information. Bands are scored based on the product of local density and IE, and the optimal number of bands is determined using a determinantal point process.
5) *MVPCA:* A variance-based ranking method derived from PCA. Bands are prioritized based on their variance in the loading factor matrix, aiding in data compression and relevant feature selection.
6) *SPMCBS:* Extends the MCBS method with subspace partitioning via correlation distance. The algorithm selects top-ranked bands from uncorrelated subspaces, reducing redundancy and improving performance.
7) *ASPSMN:* Uses adaptive subspace partitioning to divide the hyperspectral cube into subcubes. It selects the band with the lowest estimated noise from each subcube to retain unique and useful spectral information.
8) *SLFLRSC:* Performs hyperspectral BS by leveraging sample latent feature extraction, dual consistency constraints, and low-rank subspace clustering, enabling effective noise suppression, preservation of spatial–spectral information, and high-accuracy BS.

Each BS method was independently executed 30 times to ensure statistical reliability. For the proposed IMOEABS and the comparative IROABS methods, the population size was set to 30, and the maximum number of fitness evaluations was limited to 6000. To evaluate the classification performance of the selected band subsets, two widely used classifiers were employed: support vector machine (SVM) and K-nearest neighbor (KNN). In the SVM classifier, a Gaussian radial basis function (RBF) kernel was used with the penalty parameter $C = 1$ and the kernel parameter $\gamma = 1$. For the KNN classifier, the number of neighbors K was set to 8. For each dataset, 10% of the samples from each class were randomly selected as the training set, with the remaining samples used for testing. To comprehensively assess the classification performance, three evaluation metrics were adopted: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient. All empirical studies were conducted using MATLAB 2024b on PCs equipped with Intel Core i5-14 600kf CPU processors and 64-GB RAM. All multiobjective optimization algorithms were implemented and executed using the PlatEMO platform [55], a MATLAB-based framework specifically designed for evolutionary multiobjective optimization.

C. Experiment Results

The classification performance across different band subset sizes serves as a critical benchmark for evaluating the effectiveness of BS methods. Therefore, for the Salinas, Kennedy Space Center, and Botswana datasets, the average classification accuracy across 30 independent runs is reported for band subset sizes ranging from 5 to 50. In addition, classification performance at a fixed subset size of 30 is detailed at the subclass level.

1) Salinas: Fig. 6 illustrates the classification results of the IMOEABS in comparison with other methods. Under the SVM classifier, most methods, including IMOEABS, exhibit the Hughes phenomenon: as the number of selected features increases, classification accuracy declines. This is primarily due to sample sparsity in high-dimensional space, which increases model complexity, amplifies susceptibility to noise, and can lead to overfitting or unstable decision boundaries. As a result, classification accuracy using the full band set is often lower than that achieved with selected subsets. At a subset size of 5, IMOEABS achieves an OA exceeding 97%, significantly outperforming MVPCA (93%), IROABS (94%), SLFLRSC (94%), and SPMCBS (94%). This superior performance is attributed to the DSCD strategy, which penalizes the selection of adjacent, redundant bands and encourages the inclusion of spectrally diverse bands. In contrast, MVPCA relies solely on variance for band ranking and tends to select highly correlated adjacent bands, leading to redundancy.

Under the KNN classifier, IMOEABS reaches 96% OA with only five bands. As the subset size increases beyond 15, the performance of IMOEABS, EFDPC, SLFLRSC, FNGBS, and ASPSMN becomes comparable, although IMOEABS maintains a slight advantage beyond 40 bands. Due to its variance-only ranking criterion, MVPCA fails to account for spectral correlation and spatial redundancy, limiting the discriminative power of the selected features and slowing improvements in classification accuracy.

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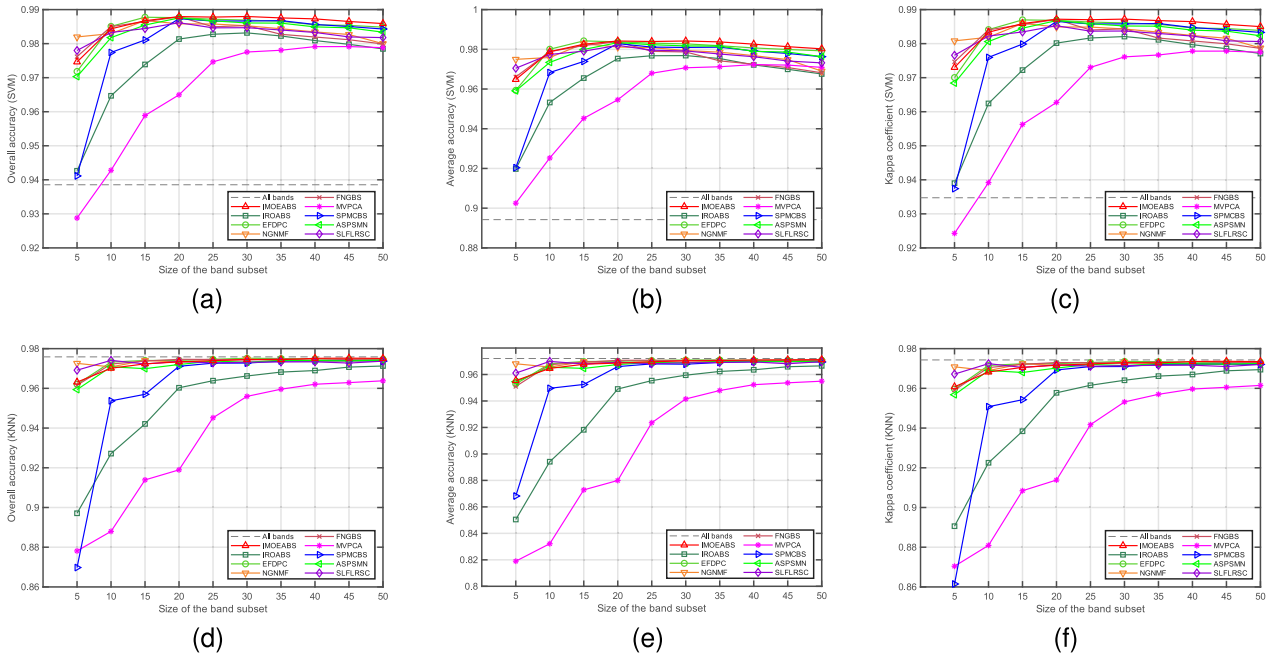


Fig. 6. Experimental results on Salinas. (a) OA curves by SVM. (b) AA curves by SVM. (c) Kappa curves by SVM. (d) OA curves by KNN. (e) AA curves by KNN. (f) Kappa curves by KNN.

TABLE I

CLASSIFICATION RESULTS OF PROPOSED IMOEABS AND OTHER STATE-OF-THE-ART METHODS ON SALINAS BY SVM (%)

Class	IMOEABS	IROABS	EFDPC	NGNMF	FNGBS	MVPCA	SPMCBS	ASPSMN	SLFLRSC
1	99.40±0.38	99.35±0.52	99.20±0.37	99.28±0.33	**99.55±0.18**	98.58±0.50	99.22±0.29	99.16±0.54	98.76±0.42
2	99.31±0.16	99.21±0.17	99.29±0.25	99.28±0.12	**99.35±0.12**	98.85±0.27	99.35±0.16	99.31±0.20	99.20±0.16
3	99.72±0.16	99.68±0.15	**99.74±0.13**	99.68±0.19	99.64±0.17	99.71±0.06	99.64±0.20	99.68±0.14	99.63±0.18
4	99.37±0.43	99.45±0.36	99.29±0.59	99.37±0.31	99.25±0.47	**99.62±0.20**	99.45±0.33	99.22±0.42	99.33±0.43
5	**98.08±0.38**	97.61±0.32	98.01±0.37	97.93±0.40	98.02±0.44	97.89±0.20	98.06±0.50	98.01±0.31	97.99±0.44
6	99.39±0.33	99.25±0.23	99.40±0.26	99.41±0.22	99.30±0.31	99.18±0.18	**99.47±0.22**	99.41±0.28	99.30±0.23
7	99.29±0.16	99.09±0.23	99.27±0.16	99.25±0.20	99.17±0.19	98.95±0.28	99.26±0.20	99.25±0.15	**99.32±0.13**
8	**99.05±0.25**	98.81±0.23	98.90±0.29	98.82±0.34	98.86±0.33	98.44±0.27	99.02±0.26	98.71±0.28	98.93±0.24
9	**99.68±0.14**	99.60±0.10	99.63±0.15	99.62±0.14	99.63±0.12	99.13±0.24	99.61±0.17	99.65±0.12	99.57±0.16
10	96.39±0.93	94.41±1.41	96.48±1.04	96.19±1.27	**96.53±1.38**	88.29±1.24	96.26±1.00	96.28±1.02	95.17±1.15
11	**97.69±1.33**	94.49±2.02	96.88±2.25	96.78±1.73	96.01±1.92	92.84±1.65	95.11±1.90	96.94±1.84	96.84±1.60
12	**99.64±0.22**	99.38±0.36	99.58±0.18	99.39±0.31	99.37±0.27	99.53±0.33	99.48±0.23	99.60±0.19	99.49±0.23
13	**97.01±1.53**	93.62±2.77	96.72±1.53	93.37±3.38	92.72±3.24	94.04±1.89	95.54±1.48	96.27±1.53	94.78±2.10
14	95.39±0.98	**95.47±1.15**	95.11±1.00	94.75±1.16	94.55±1.40	94.84±1.37	95.21±1.53	94.53±1.51	94.85±1.21
15	**98.91±0.24**	98.31±0.30	98.74±0.26	98.51±0.38	98.53±0.35	98.62±0.30	98.82±0.26	98.65±0.30	98.48±0.29
16	**96.39±0.83**	95.22±0.73	96.18±1.04	95.36±1.15	95.38±1.14	94.63±1.00	96.01±0.98	96.08±1.07	95.02±1.03
OA	**98.80±0.06**	98.32±0.11	98.69±0.05	98.52±0.10	98.53±0.09	97.75±0.05	98.66±0.05	98.61±0.07	98.47±0.05
AA	**98.42±0.14**	97.68±0.24	98.27±0.15	97.94±0.29	97.87±0.25	97.07±0.15	98.09±0.16	98.17±0.14	97.92±0.14
kappa	**98.72±0.06**	98.21±0.11	98.61±0.05	98.43±0.11	98.44±0.10	97.61±0.06	98.58±0.06	98.52±0.07	98.37±0.06

Tables I and II report the classification performance of IMOEABS and other advanced methods under both SVM and KNN classifiers. Leveraging a bi-objective fitness function (interclass distance and IE) and an adaptive operator mechanism, IMOEABS achieves an OA of 98.80% and AA of 98.42% with SVM. It also performs well in complex classes (such as 99.68% in Class 9), demonstrating a strong balance between discriminative power and redundancy control. As a swarm intelligence-based approach, IROABS attains 98.32% OA under SVM but drops to 96.62% under KNN. Its class-wise performance exhibits greater variability, suggesting sensitivity to initialization and parameter tuning. Both EFDPC and SPMCBS achieve competitive OA values under SVM (98.69% and 98.66%, respectively), indicating that clustering and multicriteria fusion strategies can yield stable results. However, SPMCBS incurs high computational costs in high-dimensional settings. FNGBS attains OA scores of 98.53% (SVM) and 97.43% (KNN) using neighborhood grouping strategies.

2) Kennedy Space Center: In Fig. 7, based on the relationship between the number of bands and classification accuracy, accuracy generally improves as the number of selected bands increases. When the number of bands in IMOEABS is small (such as 5–10), its accuracy is lower than that of other methods. However, due to the advantages of multiobjective optimization, IMOEABS quickly identifies effective bands. Once the number of bands exceeds 15, its accuracy surpasses that of competing methods. Similarly, EFDPC, which

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TABLE II
CLASSIFICATION RESULTS OF PROPOSED IMOEABS AND OTHER STATE-OF-THE-ART METHODS ON SALINAS BY KNN (%)

Class	IMOEABS	IROABS	EFDPC	NGNMF	FNGBS	MVPCA	SPMCBS	ASPSMN	SLFLRSC
1	95.55±0.90	95.51±0.47	96.32±0.53	96.53±0.43	**96.90±0.46**	94.53±0.66	96.46±0.37	95.48±1.03	94.90±0.71
2	98.93±0.23	98.17±0.40	99.05±0.12	**99.06±0.16**	99.04±0.14	94.40±0.68	98.85±0.15	98.93±0.16	98.89±0.16
3	99.75±0.10	99.01±0.79	99.79±0.03	99.79±0.03	99.79±0.05	99.50±0.26	99.79±0.02	99.79±0.03	**99.79±0.05**
4	99.60±0.13	**99.67±0.10**	99.62±0.10	99.64±0.10	99.56±0.10	99.56±0.19	99.52±0.40	99.60±0.13	99.63±0.10
5	**98.10±0.35**	96.75±0.67	98.05±0.24	97.92±0.38	97.86±0.31	97.59±0.50	98.06±0.44	97.98±0.31	97.76±0.27
6	99.00±0.37	97.68±0.62	**99.16±0.26**	99.09±0.25	99.15±0.24	97.72±0.76	99.14±0.24	98.88±0.22	98.73±0.38
7	98.95±0.13	98.77±0.19	99.03±0.09	**99.05±0.11**	98.98±0.10	97.48±0.47	98.90±0.13	99.01±0.10	99.04±0.09
8	97.23±0.15	97.14±0.16	97.18±0.20	97.09±0.17	97.01±0.19	**97.59±0.16**	97.21±0.18	97.17±0.23	97.13±0.22
9	**98.68±0.11**	98.47±0.12	98.61±0.16	98.62±0.16	98.57±0.15	98.29±0.31	98.55±0.10	98.56±0.17	98.51±0.15
10	90.28±0.93	88.50±1.09	90.60±0.87	90.71±0.91	**90.72±1.05**	79.20±1.70	89.72±0.66	90.29±1.07	90.13±0.87
11	**92.12±1.50**	91.82±1.12	91.70±1.40	91.34±0.96	91.24±1.21	88.42±2.04	91.43±1.21	91.46±1.36	91.89±0.79
12	99.88±0.15	99.72±0.23	**99.98±0.05**	99.91±0.09	99.90±0.10	99.46±0.19	99.61±0.27	99.90±0.08	99.97±0.05
13	97.34±0.37	97.41±0.36	97.44±0.44	97.28±0.69	**97.47±0.34**	97.00±0.42	97.29±0.36	97.33±0.52	97.35±0.37
14	**92.44±0.77**	91.88±1.07	91.96±1.14	92.15±1.09	91.93±1.01	86.11±2.97	91.92±1.08	92.38±0.91	92.43±0.92
15	97.98±0.37	97.29±0.49	98.08±0.36	97.99±0.36	97.73±0.34	**98.78±0.23**	97.63±0.37	97.58±0.31	97.90±0.39
16	96.73±0.93	87.39±3.31	**96.95±0.90**	96.70±1.13	96.93±1.19	80.82±2.11	94.37±1.57	95.72±1.34	96.06±1.39
OA	97.45±0.08	96.62±0.18	**97.52±0.05**	97.47±0.04	97.43±0.04	95.60±0.09	97.27±0.05	97.31±0.05	97.31±0.06
AA	97.03±0.15	95.95±0.26	**97.09±0.09**	97.05±0.11	97.05±0.09	94.15±0.18	96.78±0.11	96.88±0.11	96.88±0.10
kappa	97.29±0.08	96.41±0.19	**97.36±0.05**	97.31±0.04	97.27±0.04	95.32±0.10	97.10±0.06	97.14±0.05	97.14±0.06

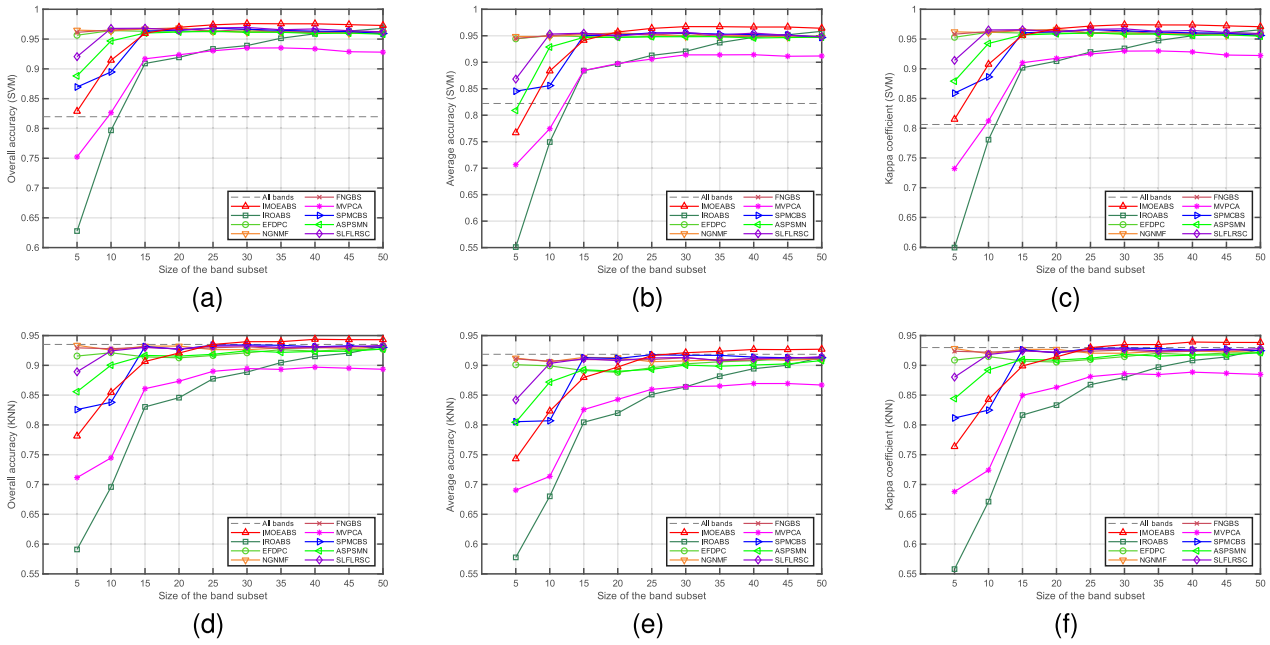


Fig. 7. Experimental results on Kennedy Space Center. (a) OA curves by SVM. (b) AA curves by SVM. (c) Kappa curves by SVM. (d) OA curves by KNN. (e) AA curves by KNN. (f) Kappa curves by KNN.

employs density-based sorting, shows a notable improvement in accuracy around ten bands. SPMCBS, leveraging multicriteria decision-making, exhibits a robust performance curve throughout. In contrast, MVPCA, which does not consider band correlation, maintains relatively low accuracy even when the number of bands increases to 30–40. This suggests its inefficacy in capturing relevant spectral information. FNGBS, utilizing neighborhood grouping, shows a steady improvement in accuracy with the increasing number of bands.

In terms of overall classification performance from Tables III and IV, IMOEABS achieves an OA of 97.60% and an AA of 96.73% under the SVM classifier. Its multiobjective optimization mechanism effectively balances band discriminability and redundancy, yielding the highest classification performance. IROABS attains an OA of 93.91% under SVM and 88.87% under KNN, with evident performance fluctuations, reflecting issues related to randomness and parameter sensitivity. EFDPC records an OA of 96.37% under SVM and 92.10% under KNN, exhibiting stable yet slightly inferior performance compared to IMOEABS. FNGBS achieves an OA of 96.65% under SVM and 93.11% under KNN, demonstrating consistent stability. MVPCA, which neglects interband correlation, results in the lowest accuracy, with 93.49% OA under SVM and 89.46% under KNN. SPMCBS performs robustly, achieving 96.65% OA under SVM and 93.42% under KNN, while ASPSMN yields 96.15% and 92.44%, respectively, maintaining general stability but slightly lower performance.

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TABLE III
CLASSIFICATION RESULTS OF PROPOSED IMOEABS AND OTHER STATE-OF-THE-ART METHODS ON KENNEDY SPACE CENTER BY SVM (%)

Class	IMOEABS	IROABS	EFDPC	NGNMF	FNGBS	MVPCA	SPMCBS	ASPSMN	SLFLRSC
1	99.68±0.54	98.80±1.32	**100.00±0.00**	99.97±0.12	99.94±0.14	98.57±1.36	99.90±0.21	99.95±0.13	99.92±0.18
2	95.04±2.53	92.02±3.15	95.71±1.60	95.11±2.25	95.16±2.16	93.53±3.12	95.31±1.96	**96.29±1.51**	95.88±1.66
3	**95.96±2.50**	92.51±4.15	94.13±3.35	92.36±4.91	91.97±3.65	94.03±3.60	93.28±3.22	91.57±4.00	93.78±3.65
4	**96.48±2.10**	93.63±2.88	89.78±5.08	92.82±3.39	94.80±3.43	92.53±2.69	94.16±3.47	93.14±4.64	94.11±3.38
5	**89.47±4.85**	76.71±5.77	87.03±5.87	84.37±4.36	86.21±5.14	73.75±6.08	84.55±5.99	85.13±7.07	85.56±6.26
6	**95.45±3.31**	84.95±3.84	91.94±5.63	92.22±4.67	94.11±4.64	84.32±4.91	95.24±4.57	93.38±5.00	90.18±5.44
7	99.01±0.92	97.87±1.83	99.26±3.30	**99.57±0.60**	99.54±0.67	94.15±6.30	99.57±1.10	97.20±6.40	99.40±0.54
8	97.35±1.38	89.59±4.52	97.43±1.66	96.65±1.06	97.34±1.78	93.08±3.24	97.74±0.94	97.04±1.02	**98.08±1.15**
9	98.83±0.77	97.93±1.28	98.51±1.50	99.00±1.06	99.14±0.93	96.84±2.03	**99.34±0.69**	99.06±1.04	99.30±0.90
10	**97.21±1.60**	90.90±3.22	93.41±3.16	93.26±3.14	93.39±3.18	88.24±4.00	94.42±2.79	92.46±2.91	95.88±2.87
11	**95.48±2.20**	91.96±3.47	93.40±2.60	91.34±2.81	92.59±3.07	92.02±3.58	91.56±3.64	91.14±3.96	93.22±3.42
12	98.20±1.11	90.44±3.29	96.29±3.45	98.10±1.98	98.37±1.71	87.46±4.44	97.75±2.09	97.90±2.42	**98.83±2.13**
13	99.34±0.73	99.45±0.28	99.34±0.93	98.49±0.86	98.85±0.92	**99.59±0.41**	98.75±0.80	98.45±1.04	99.14±0.90
OA	**97.60±0.24**	93.91±0.90	96.37±0.39	96.16±0.30	96.65±0.22	93.49±0.43	96.65±0.25	96.15±0.47	96.96±0.21
AA	**96.73±0.50**	92.06±1.12	95.09±0.60	94.87±0.42	95.49±0.35	91.39±0.76	95.51±0.50	94.82±0.74	95.64±0.46
kappa	**97.41±0.26**	93.42±0.97	96.07±0.42	95.85±0.33	96.38±0.24	92.97±0.47	96.38±0.27	95.84±0.51	96.71±0.23

TABLE IV
CLASSIFICATION RESULTS OF PROPOSED IMOEABS AND OTHER STATE-OF-THE-ART METHODS ON KENNEDY SPACE CENTER BY KNN (%)

Class	IMOEABS	IROABS	EFDPC	NGNMF	FNGBS	MVPCA	SPMCBS	ASPSMN	SLFLRSC
1	99.68±0.27	98.73±1.05	99.98±0.06	**100.00±0.00**	**100.00±0.00**	99.21±0.69	100.00±0.03	99.71±0.21	99.96±0.12
2	92.30±1.99	90.03±2.31	95.77±0.57	94.16±1.48	94.25±1.47	91.48±2.12	94.34±1.45	**96.12±0.68**	94.46±1.18
3	93.43±3.23	89.00±3.93	97.30±3.59	96.52±2.02	95.74±2.66	89.64±3.39	**97.90±1.93**	95.45±2.41	96.94±3.63
4	**93.13±3.02**	91.10±3.02	80.66±7.15	85.67±7.49	90.44±5.76	89.40±3.14	91.01±4.84	89.79±4.77	86.40±7.33
5	**74.09±5.86**	64.57±5.24	68.90±9.07	70.64±4.54	70.51±5.99	61.82±7.67	70.07±5.57	69.63±5.75	69.17±6.31
6	85.24±5.96	75.60±6.71	86.10±3.45	87.06±4.32	86.39±4.66	73.06±4.25	**87.62±5.68**	82.51±3.84	84.13±3.21
7	98.97±0.86	98.55±1.29	**99.65±0.90**	98.40±3.18	98.79±3.22	91.56±6.70	99.08±3.51	89.15±8.73	99.40±0.87
8	88.32±3.73	71.25±4.32	89.05±2.95	88.67±2.36	90.22±1.59	79.25±3.38	89.70±3.32	**91.14±2.13**	90.92±2.55
9	98.92±0.71	97.21±1.31	98.90±0.95	99.42±0.53	99.44±0.47	97.34±1.86	**99.54±0.47**	99.53±0.45	99.05±0.73
10	**91.51±3.12**	81.32±4.43	89.10±2.59	87.11±3.65	88.35±3.15	89.12±2.65	86.51±5.29	88.06±3.42	91.00±3.35
11	92.15±2.18	89.40±3.25	**93.45±1.99**	91.44±1.76	91.18±1.60	86.73±3.29	91.19±1.65	92.10±1.36	93.09±1.82
12	**91.58±3.59**	77.17±5.32	76.62±5.65	82.61±4.17	83.26±3.70	77.17±6.34	87.23±5.50	79.32±3.32	84.44±4.89
13	98.19±0.65	**99.16±0.50**	97.75±0.40	97.94±0.41	98.21±0.84	97.93±0.72	97.76±0.11	98.03±0.56	97.91±0.52
OA	**93.97±0.39**	88.87±1.07	92.10±0.43	92.63±0.34	93.11±0.37	89.46±0.53	93.42±0.62	92.44±0.38	93.29±0.31
AA	**92.12±0.70**	86.39±1.41	90.25±0.69	90.74±0.57	91.29±0.56	86.44±0.99	91.69±0.76	90.04±0.71	91.30±0.53
kappa	**93.48±0.42**	87.97±1.16	91.47±0.46	92.03±0.37	92.56±0.40	88.61±0.57	92.89±0.67	91.83±0.41	92.75±0.34

3) Botswana: Fig. 8 illustrates the classification results of the IMOEABS in comparison with other methods. The relationship between the number of selected bands and classification accuracy under the SVM classifier indicates that all methods exhibit improved performance as the number of bands increases. Leveraging multiobjective optimization, IMOEABS achieves competitive accuracy and is only slightly outperformed by EFDPC when the number of bands is small (such as 5). EFDPC, which is based on density ranking, shows a significant increase in accuracy around ten bands. SPMCBS, utilizing multicriteria decision-making, demonstrates a consistently robust performance curve. In contrast, MVPCA, due to its disregard for band correlation, maintains relatively low accuracy even with 30–40 bands, highlighting the inefficiency of traditional unsupervised methods. FNGBS, which applies neighborhood grouping, exhibits steadily increasing accuracy, although its initial growth rate is lower than that of IMOEABS. Under the KNN classifier, a similar trend is observed. IMOEABS maintains strong performance across varying band sizes (5–50). As a swarm intelligence-based method, IROABS yields moderate results on the Botswana dataset. Both EFDPC and SPMCBS achieve high accuracy around 15–20 bands, demonstrating effective band screening capabilities. FNGBS is slightly weaker than EFDPC in the early stage, while MVPCA continues to show poor performance even with 40–50 bands, again confirming its limitation in handling interband correlation. In summary, modern optimization methods (IMOEABS) achieve high classification accuracy with fewer bands, whereas traditional methods are less efficient, underscoring the advantage of advanced strategies in BS.

From Tables V and VI, utilizing a multiobjective optimization strategy, IMOEABS achieves an OA of 91.49% under SVM and 87.42% under KNN, effectively balancing band IE and redundancy. This validates the effectiveness of multiobjective strategies in hyperspectral BS. IROABS, constrained by randomness and parameter sensitivity, achieves an OA of 81.41% under SVM and 75.46% under KNN. EFDPC, based on density sorting, achieves 91.25% OA under SVM, which is comparable to IMOEABS, reflecting the efficiency of density-based screening. Supported by neighborhood grouping, FNGBS obtains 90.65% OA under SVM and 84.42% under KNN, showing stable but slightly lower accuracy. The traditional method MVPCA, which neglects interband correlation, yields the lowest accuracy: 80.28% under SVM and 75.86% under KNN. SPMCBS, via multicriteria decision-making, achieves 90.63% and 84.16% under SVM and KNN,

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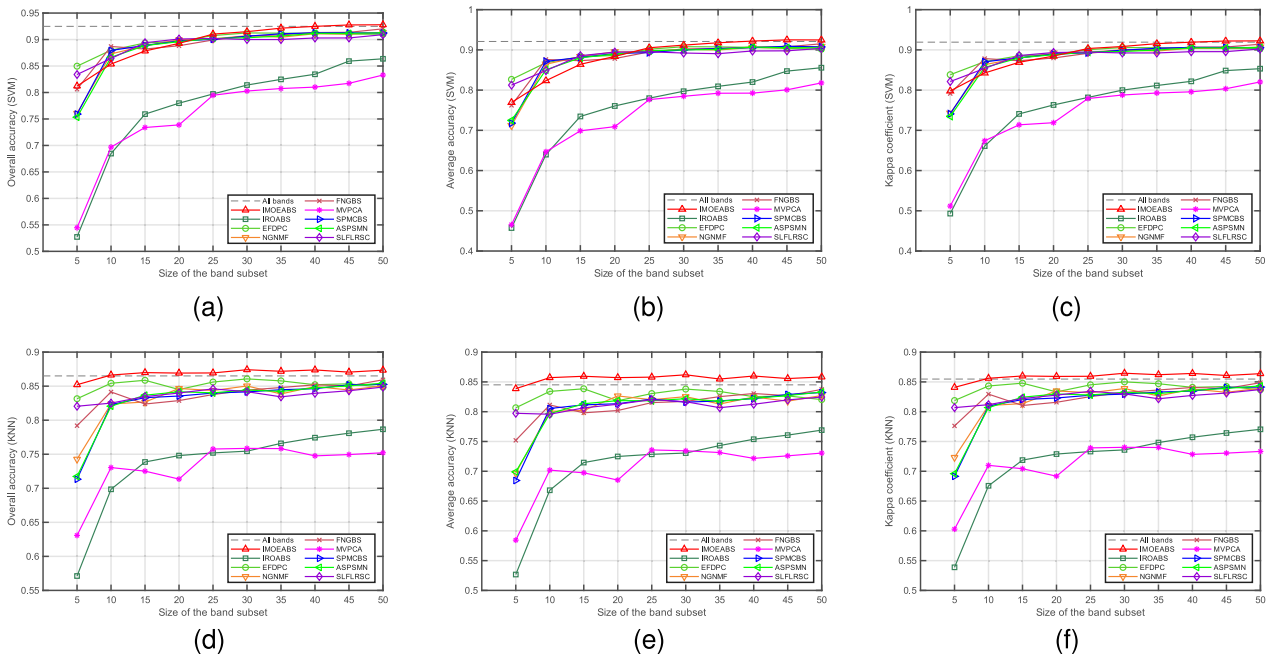


Fig. 8. Experimental results on Botswana. (a) OA curves by SVM. (b) AA curves by SVM. (c) Kappa curves by SVM. (d) OA curves by KNN. (e) AA curves by KNN. (f) Kappa curves by KNN.

TABLE V
CLASSIFICATION RESULTS OF PROPOSED IMOEABS AND OTHER STATE-OF-THE-ART METHODS ON BOTSWANA BY SVM (%)

Class	IMOEABS	IROABS	EFDPC	NGNMF	FNGBS	MVPCA	SPMCBS	ASPSMN	SLFLRSC
1	94.84±2.53	91.92±1.90	94.68±3.13	94.54±3.53	95.03±3.20	91.69±2.35	94.10±2.48	94.66±3.14	**95.45±2.64**
2	75.86±9.22	62.27±12.12	75.49±10.12	75.31±8.16	74.51±10.98	57.47±9.29	69.89±12.28	**78.02±11.00**	67.80±10.61
3	97.61±2.48	96.71±2.41	97.29±2.68	**97.77±1.42**	96.65±2.48	94.40±4.76	97.37±2.26	96.52±2.67	95.65±4.02
4	**92.49±2.57**	81.61±3.22	91.47±3.29	91.99±3.92	90.83±3.68	81.90±3.75	92.07±3.93	91.88±4.68	90.60±3.91
5	85.48±2.89	82.85±4.24	85.72±3.45	**86.27±3.87**	83.87±3.81	83.73±4.66	84.94±3.90	84.88±3.71	84.12±3.65
6	**82.22±4.01**	72.45±4.20	80.65±4.76	77.67±4.87	78.09±5.62	71.42±3.98	79.94±5.54	77.73±4.36	78.58±4.91
7	92.25±3.36	88.08±4.09	93.39±3.46	93.05±5.73	**93.53±3.22**	87.83±4.19	91.36±4.72	92.52±4.22	93.48±3.42
8	**96.87±2.57**	63.93±5.03	96.14±3.95	96.28±2.70	94.77±3.70	60.27±5.22	94.59±4.25	95.39±3.83	94.85±5.48
9	94.96±2.04	80.67±3.90	94.99±2.22	94.44±2.78	94.86±2.61	77.29±4.73	**96.53±1.36**	94.19±1.71	92.80±3.31
10	83.98±3.40	72.69±5.33	82.90±4.50	81.87±5.25	83.38±5.99	75.93±4.84	82.77±5.34	82.05±5.68	**84.02±6.15**
11	90.90±2.69	87.66±3.12	91.93±2.59	91.68±4.01	**92.29±3.18**	85.52±2.24	91.69±3.17	91.12±3.23	91.22±3.33
12	95.32±3.40	70.49±4.39	**97.06±1.92**	96.44±2.71	95.40±2.57	70.96±4.50	95.73±2.37	95.60±2.54	95.34±3.26
13	**97.11±1.82**	89.11±3.10	96.40±2.22	95.20±2.89	95.55±2.59	86.32±3.12	94.66±4.42	94.11±3.79	94.48±3.05
14	**96.67±3.39**	75.61±10.97	92.90±6.96	90.31±7.65	94.08±7.04	73.80±11.35	94.31±4.49	91.25±7.52	90.67±8.44
OA	**91.49±0.50**	81.41±1.34	91.25±0.31	90.72±0.60	90.65±0.54	80.28±0.61	90.63±0.35	90.33±0.53	90.00±0.42
AA	**91.18±0.65**	79.72±1.63	90.79±0.62	90.20±0.78	90.20±0.87	78.47±0.85	90.00±0.68	89.99±0.86	89.22±0.65
kappa	**90.84±0.54**	79.99±1.44	90.59±0.34	90.02±0.64	89.94±0.58	78.78±0.66	89.92±0.38	89.60±0.57	89.24±0.45

respectively, with robust performance. ASPSMN records OA values of 90.33% and 84.44%, respectively, showing slightly lower overall stability. In terms of subcategory classification, IMOEABS achieves the best performance in 8 subcategories.

VI. DISCUSSION AND ANALYSES

To comprehensively evaluate the performance of IMOEABS, this section presents a multifaceted analysis of the experimental results, including algorithm performance evaluation, improvement effectiveness, band subset composition, performance under small training sets, and the computational efficiency of different BS methods.

A. Algorithm Performance Testing of IMOEABS

To verify the optimization capability of the IMOEABS method, several classical multiobjective algorithms are selected for comparison, including MORINGPSOSCD [49], MOEADEA [56], NSGAII [44], and NSGAIII [57]. According to the data tables, the IMOEABS algorithm exhibits significant advantages due to its bi-objective optimization strategy—maximizing IE and the MBD—along with the incorporation of DSCD and a dynamic search strategy based on entropy-driven selection. The experimental results are shown in Table VII.

On the Salinas dataset, IMOEABS achieves the highest IE of 8.346 and an MBD of 192.6766, resulting in an OA of 98.72%. These results confirm the effectiveness of the bi-objective function in improving interclass separability and information richness. For the Kennedy Space Center dataset, its MBD reaches 94.7322, substantially higher than that of the comparison algorithms, and the OA is 96.82%, highlighting the pivotal role of interclass separability in classification.

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TABLE VI

CLASSIFICATION RESULTS OF PROPOSED IMOEABS AND OTHER STATE-OF-THE-ART METHODS ON BOTSWANA BY KNN (%)

Class	IMOEABS	IROABS	EFDPC	NGNMF	FNGBS	MVPCA	SPMCBS	ASPSMN	SLFLRSC
1	**92.46±1.62**	86.91±2.92	91.95±0.91	92.19±1.38	92.32±1.39	88.33±2.53	92.26±1.67	91.28±2.12	91.44±1.22
2	63.19±12.23	59.08±12.20	49.63±9.37	46.26±8.46	41.14±9.34	**64.51±9.13**	34.62±9.37	45.71±8.40	38.97±10.90
3	**99.88±0.26**	98.38±1.90	99.10±0.43	98.20±0.33	98.20±0.59	98.30±2.75	98.73±0.38	98.35±0.68	97.49±1.06
4	**91.61±2.92**	79.53±3.29	88.62±2.79	88.48±2.86	87.25±2.75	80.88±3.70	88.77±2.02	87.70±1.57	88.39±2.44
5	**79.57±3.70**	72.84±5.68	78.13±3.00	77.45±3.60	77.64±3.64	74.33±5.72	77.18±3.61	75.06±4.57	75.39±3.51
6	**68.25±4.68**	52.51±6.92	64.66±5.24	63.68±5.38	62.99±5.52	52.25±5.49	64.21±4.38	64.89±5.53	64.34±4.44
7	89.59±4.53	83.58±5.13	**91.79±3.06**	90.26±2.91	90.30±2.66	83.78±4.45	89.03±3.09	88.73±3.09	89.87±3.17
8	91.95±5.43	60.73±6.93	92.68±4.93	93.33±6.59	**92.79±4.86**	59.87±7.35	89.96±5.05	92.13±3.96	93.88±2.88
9	90.68±2.07	71.99±4.89	**91.53±1.78**	87.44±3.01	87.51±3.29	73.04±4.20	86.02±3.32	88.30±3.63	85.65±3.18
10	**85.19±3.33**	72.15±7.64	82.91±5.34	84.44±5.36	82.03±6.13	74.19±5.95	83.06±4.90	81.30±6.71	81.42±4.99
11	88.11±2.71	83.25±7.24	**88.99±4.12**	87.82±4.37	88.63±5.08	84.56±4.03	87.19±5.12	88.21±4.48	89.72±4.14
12	93.50±3.78	73.72±5.95	**94.09±3.06**	93.42±1.97	90.92±2.45	75.87±6.77	90.88±2.75	91.00±2.82	93.15±2.26
13	**94.65±2.48**	85.26±5.21	93.55±2.33	92.84±2.48	91.60±1.67	80.80±6.80	91.37±2.40	91.60±2.10	89.59±2.71
14	**78.08±11.34**	42.86±17.56	65.53±9.50	59.53±10.71	60.86±10.51	37.41±15.25	69.14±9.50	66.27±8.45	64.12±9.84
OA	**87.42±0.66**	75.46±2.04	86.07±0.44	85.03±0.41	84.42±0.30	75.86±0.56	84.16±0.38	84.44±0.42	84.19±0.39
AA	**86.19±0.88**	73.06±2.65	83.80±0.87	82.52±0.80	81.73±0.83	73.44±1.00	81.60±0.74	82.18±0.64	81.67±0.71
kappa	**86.46±0.71**	73.59±2.20	85.01±0.47	83.89±0.44	83.24±0.32	74.02±0.61	82.95±0.41	83.25±0.46	82.98±0.42

TABLE VII

CLASSIFICATION RESULTS OF PROPOSED IMOEABS AND OTHER EAS METHODS BY SVM

Dataset	Algorithm	IE	MBD	OA	AA	kappa
Salinas	IMOEABS	**8.346 ± 0.08**	192.6766±0.0014	**98.72±0.09**	**98.05±0.19**	**98.63±0.09**
	MORingPSOSCD	8.148 ± 0.16	193.0419±0.0027	98.48±0.13	97.83±0.19	98.39±0.14
	MOEADDE	7.980 ± 0.10	**194.6943±0.0017**	98.21±0.13	97.52±0.14	98.10±0.13
	NSGAII	8.082± 0.17	194.0374±0.0028	98.34±0.20	97.64±0.27	98.24±0.21
	NSGAIII	8.112 ± 0.17	194.1229±0.0028	98.41±0.18	97.71±0.22	98.31±0.19
Kennedy Space Center	IMOEABS	**6.798± 0.35**	**94.7322±0.0059**	**96.82±0.42**	**96.15±0.64**	**96.56±0.45**
	MORingPSOSCD	4.116 ± 1.24	89.1199±0.0207	93.27±4.34	91.89±3.58	92.73±4.69
	MOEADDE	4.932 ± 1.21	90.1921±0.0202	95.05±2.57	93.83±2.44	94.65±2.78
	NSGAII	4.680 ± 1.21	90.2262±0.0202	95.59±0.81	94.22±1.08	95.24±0.87
	NSGAIII	4.674 ± 1.13	90.1883±0.0189	95.34±0.99	93.86±1.25	94.96±1.07
Botswana	IMOEABS	**8.486 ± 0.03**	132.5719±0.0004	**90.69±0.95**	**90.22±1.02**	**89.98±1.02**
	MORingPSOSCD	8.427 ± 0.05	134.2735±0.0008	89.26±0.76	87.79±1.26	88.44±0.82
	MOEADDE	8.385 ± 0.05	**135.9435±0.0008**	88.80±0.63	86.82±1.08	87.94±0.67
	NSGAII	8.446 ± 0.06	135.0996±0.0010	89.16±1.21	87.74±1.68	88.33±1.31
	NSGAIII	8.450 ± 0.04	134.4308±0.0007	88.66±0.88	86.83±1.36	87.80±0.95

TABLE VIII

PERFORMANCE COMPARISON OF MULTIOBJECTIVE OPTIMIZATION ALGORITHMS ON IGD, HV, PCC, AND SSIM

Dataset	Algorithm	IGD	HV	PCC	SSIM
Salinas	IMOEABS	**10.481**	2.034	**0.283**	0.341
	MORingPSOSCD	14.896	1.573	0.316	0.337
	MOEADDE	13.151	1.906	0.309	**0.326**
	NSGAII	10.509	**2.388**	0.322	0.366
	NSGAIII	11.268	2.195	0.321	0.366
Kennedy Space Center	IMOEABS	**85.511**	**1.461**	0.438	**0.955**
	MORingPSOSCD	92.238	1.378	0.438	0.959
	MOEADDE	90.141	1.420	**0.387**	0.954
	NSGAII	85.588	1.434	0.695	0.984
	NSGAIII	86.682	1.438	0.634	0.980
Botswana	IMOEABS	**71.111**	1.303	**0.899**	**0.539**
	MORingPSOSCD	75.761	1.240	0.907	0.559
	MOEADDE	72.340	**1.362**	0.913	0.593
	NSGAII	71.240	1.307	0.924	0.628
	NSGAIII	71.751	1.296	0.918	0.615

On the Botswana dataset, IMOEABS obtains the highest IE (8.486); although its MBD is slightly lower than others, it still achieves an OA of 90.69%, emphasizing the value of IE in mitigating redundancy.

Furthermore, the algorithm demonstrates low standard deviations across datasets, indicating strong robustness. Even in complex scenarios, it effectively avoids redundancy among adjacent bands through the DSCD mechanism and balances exploration and exploitation via the dynamic search strategy. Overall, IMOEABS outperforms MORINGPSOSCD, MOEADEA, and other comparative algorithms in terms of comprehensive performance.

In addition, the effectiveness of the cost function is evident in the observed relationships among IE, MBD, and OA. For instance, on the Kennedy Space Center dataset, IMOEABS yields the highest IE and MBD, clearly outperforming all other methods. In contrast, MORINGPSOSCD records the lowest IE and MBD values, along with the lowest OA. For NSGAII, although its IE is lower than that of MOEADEA, it achieves a higher MBD.

Table VIII compares the performance of different EAs on three hyperspectral datasets in terms of convergence (IGD), diversity (HV), and subset correlation [Pearson correlation coefficient (PCC) and structural similarity index (SSIM)]. When calculating IGD and HV, the PF is obtained by running each method multiple times on each dataset. Overall, IMOEABS demonstrates competitive or superior performance. For example, on the Salinas dataset, IMOEABS achieves the lowest IGD (10.481) and a high HV (2.034), indicating

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TABLE IX
CLASSIFICATION RESULTS OF ABLATION EXPERIMENT BY SVM

Dataset	Algorithm	IE	MBD	OA	AA	kappa
Salinas	IMOEABS	**8.3520 ± 0.10**	192.9778±0.0016	**98.71±0.08**	98.01±0.18	**98.63±0.09**
	IMOEACD	8.3400 ± 0.13	191.2035±0.0021	98.41±0.57	97.56±1.11	98.31±0.61
	IMOEAQI	8.3640 ± 0.14	190.0452±0.0023	98.28±0.67	97.29±1.20	98.17±0.71
	IMOEAOURE	8.2200 ± 0.18	**193.0389±0.0030**	98.47±0.14	97.74±0.20	98.37±0.15
	IMOEAPB	8.2800 ± 0.14	192.7257±0.0023	98.56±0.08	97.83±0.18	98.47±0.08
	IMOEASCD	8.1786 ± 0.28	192.2327±0.0028	98.41±0.20	97.82±0.35	98.31±0.21
	IMOEAVCM	8.3434 ± 0.12	191.7902±0.0021	98.56±0.15	**98.02±0.28**	98.46±0.16
Kennedy Space Center	IMOEABS	**6.8520 ± 0.32**	**94.8150±0.0054**	**96.93±0.35**	**96.33±0.46**	**96.69±0.38**
	IMOEACD	6.0060 ± 1.07	91.8428±0.0178	96.12±0.46	94.96±0.81	95.80±0.50
	IMOEAQI	6.2160 ± 0.73	91.2728±0.0121	96.10±0.46	94.91±0.79	95.78±0.49
	IMOEAOURE	5.2080 ± 1.37	90.6731±0.0229	95.28±1.52	93.92±1.46	94.89±1.64
	IMOEAPB	5.4240 ± 1.13	90.1703±0.0189	95.68±0.97	94.50±1.08	95.33±1.05
	IMOEASCD	5.4573 ± 0.52	90.6900±0.0087	96.41±0.34	95.40±0.59	96.12±0.37
	IMOEAVCM	6.7332 ± 0.35	93.1977±0.0059	96.63±0.49	95.54±0.75	96.35±0.53
Botswana	IMOEABS	**8.4864 ± 0.03**	132.5719±0.0004	**90.69±0.95**	**90.22±1.02**	**89.98±1.02**
	IMOEACD	8.4415 ± 0.05	133.5586±0.0008	89.28±1.16	88.06±1.65	88.46±1.25
	IMOEAQI	8.4502 ± 0.10	132.1896±0.0016	89.35±1.25	88.37±1.46	88.53±1.34
	IMOEAOURE	8.4318 ± 0.06	133.3807±0.0010	90.24±1.29	89.39±1.73	89.50±1.39
	IMOEAPB	8.4352 ± 0.04	**135.1335±0.0007**	88.88±0.67	87.07±1.09	88.03±0.72
	IMOEASCD	8.4334 ± 0.06	134.0758±0.0011	89.26±1.02	88.02±1.49	88.44±1.10
	IMOEAVCM	8.4717 ± 0.05	132.5603±0.0008	89.93±1.20	89.33±1.34	89.16±1.29

stable convergence and a diverse set of solutions. Regarding subset correlation, IMOEABS exhibits low PCC values across datasets, indicating reduced linear redundancy, and moderate SSIM values, reflecting balanced structural similarity. In contrast, other methods generally have higher correlation values, suggesting greater redundancy. These results indicate that IMOEABS improves convergence and diversity while reducing redundancy, leading to more effective BS.

B. Ablation Experiment

To validate the effectiveness of the symmetric vertical spectral feature coding mechanism, the entropy-driven operator, and the DSCD, a series of ablation experiments is conducted. Specifically, IMOEACD replaces the DSCD-based nondominated sorting in the original IMOEABS with the traditional crowding-distance-based sorting. Meanwhile, IMOEAQI, IMOEAOURE, and IMOEAPB employ the QI operator, the Ornstein–Uhlenbeck random evolution operator, and the *DE/current-to-pbest/1* operator, respectively. All three variants retain the DSCD-based sorting mechanism. Moreover, IMOEASCD substitutes the DSCD-based sorting with an SCD-based one, while IMOEAVCM replaces the symmetric vertical spectral feature coding mechanism with the vertical spectral feature coding mechanism.

The ablation results (Table IX) highlight the contributions of each component in the IMOEABS method. On the Salinas dataset, replacing the DSCD-based sorting mechanism with the traditional crowding distance (IMOEACD) decreases the OA from 98.71% to 98.41%, while using static SCD instead of DSCD (IMOEASCD) also yields an OA of 98.41%. These observations indicate that DSCD effectively regulates decision space diversity by penalizing adjacent BS and rewarding band combinations with larger index intervals, thereby enhancing interclass separability more effectively than static SCD or conventional crowding distance.

When evaluating only the mutation operators, IMOEAOURE, which employs the Ornstein–Uhlenbeck operator, achieves an OA of 98.56% on Salinas and 95.68% on Kennedy Space Center. In comparison, IMOEAPB, which uses the DE/current-to-pbest/1 operator, achieves slightly lower OA values, confirming that the entropy-driven operator selection mechanism dynamically balances global exploration and local exploitation to produce higher quality band subsets.

Regarding the coding mechanism, replacing the symmetric vertical spectral feature coding mechanism with the traditional vertical spectral feature coding mechanism (IMOEAVCM) results in a slight decrease in OA on Salinas, while AA and Kappa remain similar. The symmetric vertical spectral feature coding mechanism maps band indices to a symmetric interval during evolution, mitigating boundary violations and preventing repeated selection of the same band. In contrast, the traditional vertical spectral feature coding mechanism uses the original interval, which is prone to asymmetry-induced boundary clipping and reduced diversity. This demonstrates that the symmetric mapping improves search stability and robustness, particularly in complex datasets such as the Kennedy Space Center and Botswana.

Overall, the DSCD-based sorting mechanism significantly improves classification accuracy by promoting diverse BS in the decision space. The entropy-driven operator selection mechanism adaptively chooses appropriate evolutionary operators to balance global and local search. Finally, the symmetric vertical spectral feature coding mechanism stabilizes the search process and prevents repeated selection of bands. The integration of these three innovations in IMOEABS leads to the most accurate and robust hyperspectral BS performance across all tested datasets.

C. Subset Analysis

1) Subset Component Analysis: Band subset composition is a crucial aspect in evaluating the performance of BS methods.

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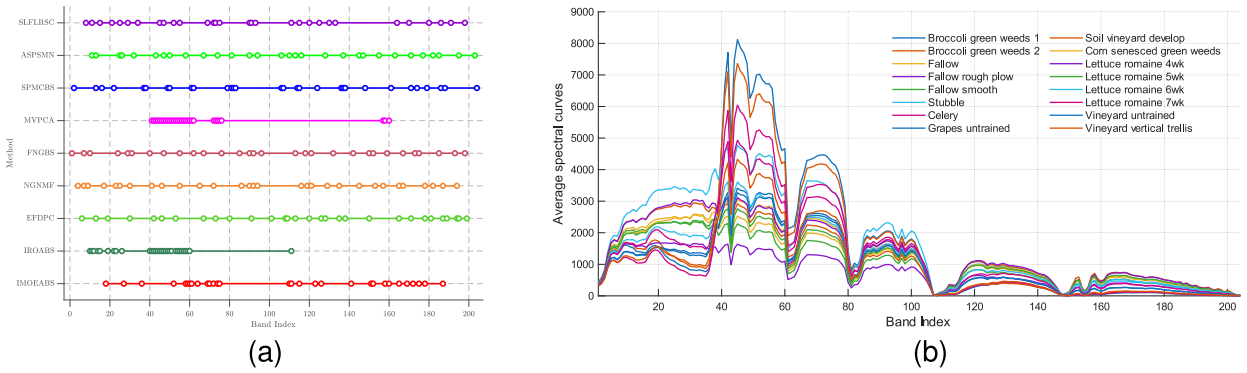


Fig. 9. Distributions of band subsets selected by different methods on Salinas. (a) Distributions of band subset. (b) Spectral curves.

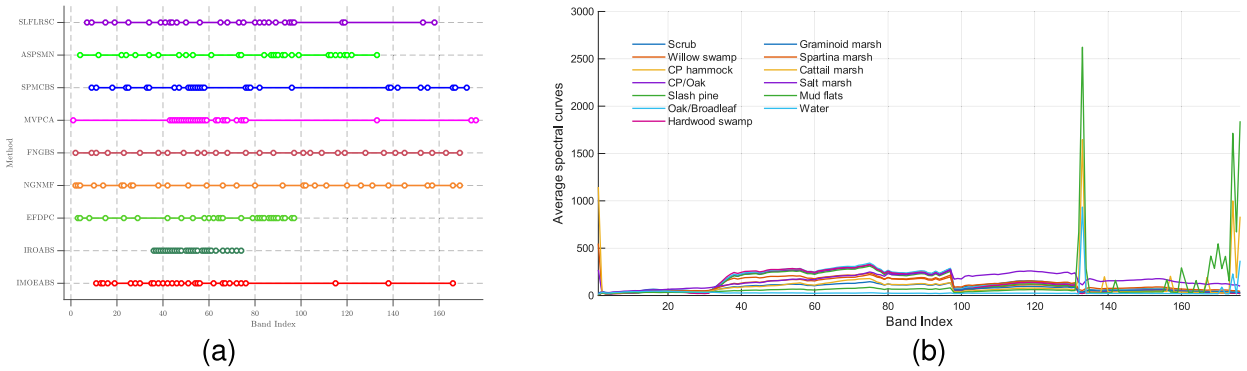


Fig. 10. Distributions of band subsets selected by different methods on Kennedy Space Center. (a) Distributions of band subset. (b) Spectral curves.

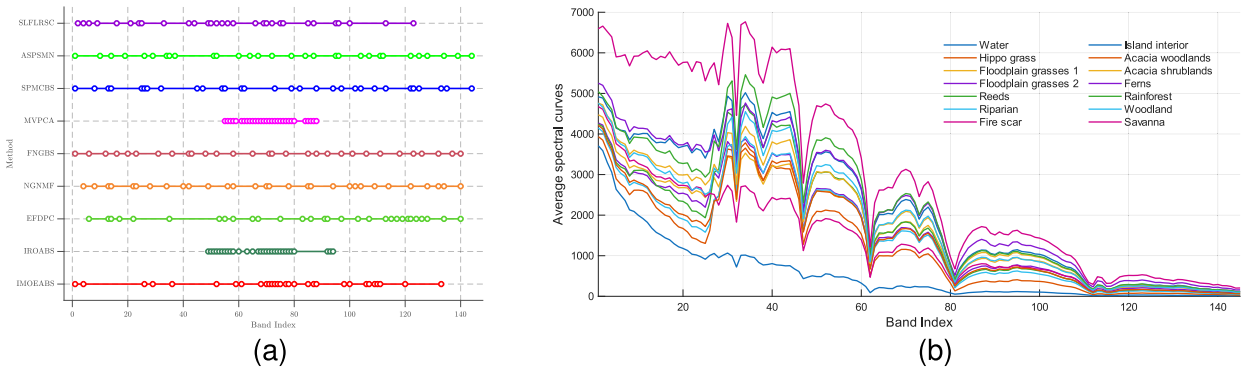


Fig. 11. Distributions of band subsets selected by different methods in Botswana. (a) Distributions of band subset. (b) Spectral curves.

Figs. 9–11 illustrate the distribution of bands selected by various algorithms across the three datasets, along with the corresponding spectral profiles. For consistency, the subset size is fixed at 30 for the Salinas, Kennedy Space Center, and Botswana datasets.

In the Salinas dataset (Fig. 9), the selected bands by IMOEABS are primarily concentrated in the range of bands 40–80, which correspond to regions with large interclass spectral differences. The algorithm effectively prioritizes bands with high spectral discriminability and low redundancy. In contrast, IROABS and MVPCA tend to select bands that are overly concentrated in high-energy regions, leading to redundancy due to dense selection. In the Kennedy Space Center dataset (Fig. 10), IMOEABS adaptively adjusts its search strategy using an entropy-driven operator, focusing on spectral regions with significant interclass separation (notably bands 20–100). Unlike IROABS, which exhibits more centralized BS, IMOEABS achieves a more balanced distribution, contributing to improved classification performance. For the Botswana dataset (Fig. 11), the selected band subset of IMOEABS is more dispersed, reflecting its ability to mitigate redundancy in complex spectral scenes. This spatial distribution of bands facilitates better class separability. Overall, the BS strategy employed by IMOEABS demonstrates clear advantages across diverse datasets, confirming its effectiveness in selecting informative and nonredundant bands under varying spectral conditions.

2) Subset Correlation Analysis: To evaluate the redundancy and structural similarity among hyperspectral bands, two complementary similarity metrics were employed: the PCC and the SSIM.

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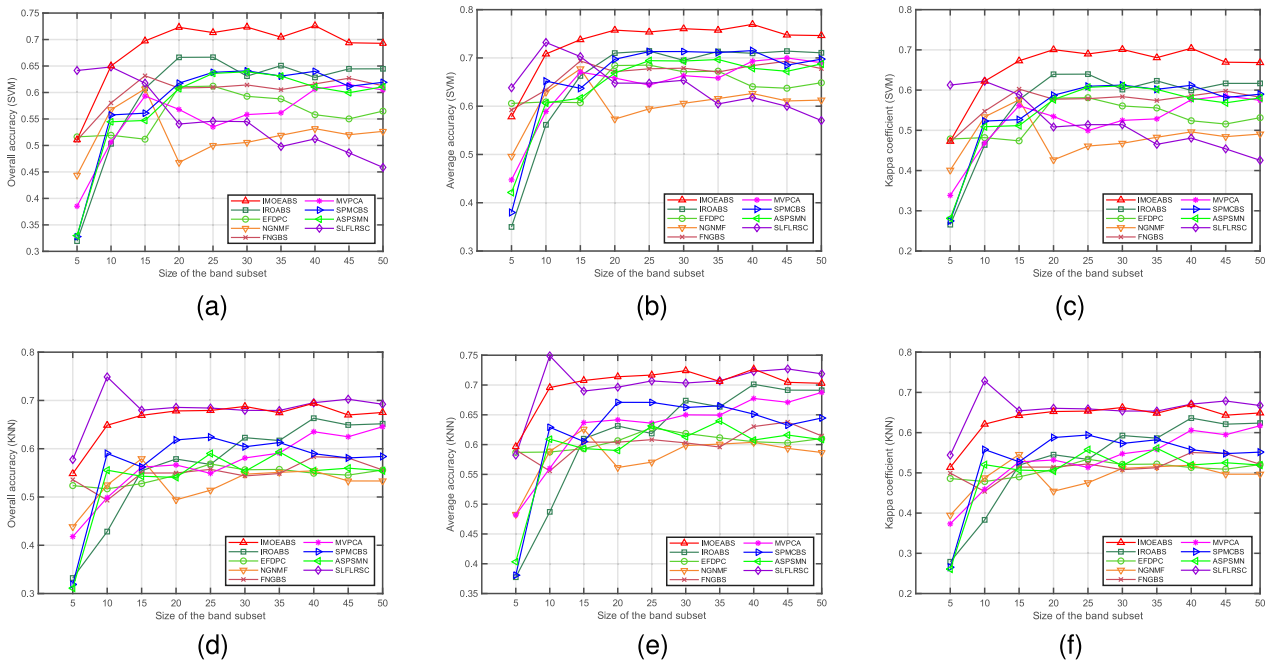


Fig. 12. Experimental results with a small sample on Kennedy Space Center. (a) OA curves by SVM. (b) AA curves by SVM. (c) Kappa curves by SVM. (d) OA curves by KNN. (e) AA curves by KNN. (f) Kappa curves by KNN.

As shown in Table X, IMOEABS achieves consistently low PCC values and moderate SSIM values across all three datasets. In particular, on the Salinas dataset, its PCC and SSIM are 0.2658 and 0.3121, respectively, which are significantly lower than those obtained by other algorithms. This indicates that IMOEABS has a distinct advantage in maintaining interband diversity. The primary improvement stems from the incorporation of the DSCD. This modification ensures that individuals are more uniformly distributed not only in the objective space during evolution but also within the band index space. Consequently, the algorithm effectively avoids the selection of redundant, closely adjacent bands, thereby enhancing the independence and discriminability of spectral features. Overall, by introducing the DSCD mechanism, IMOEABS achieves a coordinated optimization of both spectral and index space distributions, demonstrating superior performance in preserving band diversity and improving feature representation capability.

TABLE X
RESULTS OF CORRELATION RESULTS

Algorithm	Salinas		Kennedy Space Center		Botswana	
	PCC	SSIM	PCC	SSIM	PCC	SSIM
IMOEABS	**0.2658**	**0.3121**	0.4732	0.9338	0.9305	0.6578
IROABS	0.5142	0.5541	0.8559	0.9984	0.9846	0.8792
EFDPC	0.4054	0.3808	0.7179	0.9844	0.9396	0.6202
NGNMF	0.4211	0.4196	0.1908	0.9177	0.9143	0.5840
FNGBS	0.3814	0.3700	0.1547	0.8892	**0.9053**	0.5593
MVPCA	0.3077	0.3581	**0.1128**	**0.8373**	0.9068	0.6138
SPMCBS	0.3879	0.4170	0.2990	0.9143	0.9146	0.5718
ASPSMN	0.3778	0.3355	0.1940	0.9565	0.9064	**0.5574**
SLFLRSC	0.3386	0.3337	0.2995	0.9654	0.9112	0.7209

D. Small Sample Experiment

In this section, a small-sample experiment is conducted, where ten samples per subcategory are randomly selected to form the training set. Fig. 12 presents the classification results under small-sample conditions on the Kennedy Space Center dataset, with the number of selected bands varying from 5 to 50.

Table XI presents the specific classification results for different methods when the subset size is set to 30. Due to the limited training data (only ten samples per class), all methods exhibit relatively low classification accuracy. However, IMOEABS consistently achieves relatively high accuracy even under these constraints, demonstrating strong generalization capability in small-sample scenarios. Furthermore, as observed from the corresponding data table, EAs such as IMOEABS and IROABS outperform other methods, confirming their superior optimization capabilities under data-scarce conditions. SPMCBS, which incorporates multiple evaluation metrics for BS, also performs well, albeit slightly behind the evolutionary approaches. In contrast, methods based on clustering and ranking (MVPCA, EFDPC) fail to effectively identify discriminative bands due to insufficient training data, resulting in lower classification performance.

E. Calculation Time Analysis

Table XII reports the runtime of various BS methods on multiple datasets for subset sizes of 10, 20, and 30. IMOEABS, based on a MOEA, achieves superior convergence and classification performance, although it requires more computational time compared with lightweight clustering or unsupervised methods. The increased runtime arises from iterative population evolution, repeated fitness evaluations, and extensive matrix operations intrinsic to evolutionary optimization.

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TABLE XI
CLASSIFICATION ACCURACY WITH SMALL SAMPLE ON KENNEDY SPACE CENTER (%)

method	KNN			SVM		
	OA	AA	kappa	OA	AA	kappa
IMOEABS	**68.76±2.19**	**72.41±1.88**	**66.24±2.36**	**72.35±4.48**	**76.05±3.33**	**70.11±4.83**
IROABS	62.29±4.60	67.38±3.52	59.27±4.94	63.09±6.39	69.49±6.08	60.14±6.88
EFDPC	55.64±2.73	61.87±2.45	52.10±2.92	59.28±3.71	67.13±2.99	56.06±3.96
NGNMF	54.73±2.46	59.85±2.09	51.11±2.62	50.57±3.30	60.60±3.18	46.77±3.53
FNGBS	54.37±1.93	60.29±1.91	50.79±2.05	61.42±3.63	67.88±3.31	58.38±3.88
MVPCA	58.09±3.20	64.99±2.09	54.75±3.43	55.83±4.97	66.33±2.88	52.48±5.24
SPMCBS	60.41±2.44	66.23±1.88	57.31±2.60	64.09±3.85	71.32±3.27	61.25±4.12
ASPSMN	55.30±3.04	61.32±2.92	51.83±3.26	63.92±4.96	69.36±4.51	61.06±5.32
SLFLRSC	67.95±2.05	70.34±1.55	65.38±2.20	54.50±6.22	65.37±5.07	51.39±6.53

TABLE XII
CALCULATION TIME OF DIFFERENT METHODS ON DIFFERENT DATASETS (S)

Dataset	band	IMOEABS	BIMOEABS	IROABS	EFDPC	NGNMF	FNGBS	MVPCA	SPMCBS	ASPSMN	SLFLRSC
Salinas	10	0.9324	1.5988	1.0636	0.0658	0.2708	0.2588	0.0948	0.4569	0.2424	3.3433
	20	1.1565	1.5734	1.0458	0.0709	0.2762	0.2548	0.0963	0.4580	0.2387	3.3746
	30	1.3340	1.5793	1.0036	0.0664	0.2650	0.2472	0.0904	0.4534	0.2266	3.4680
Kennedy Space Center	10	0.8537	1.5427	0.9821	0.1455	0.5689	0.5038	0.4922	1.4552	0.6123	4.5739
	20	1.1324	1.5286	0.8553	0.1437	0.6058	0.4999	0.4850	1.4380	0.6048	4.5485
	30	1.3208	1.5329	0.8401	0.1425	0.6080	0.4960	0.4921	1.4524	0.6041	4.5421
Botswana	10	0.7476	1.4325	0.6663	0.1442	0.5937	0.4869	0.3825	1.4526	0.6495	4.3607
	20	1.0182	1.4206	0.6809	0.1468	0.6297	0.5064	0.3985	1.4323	0.6503	4.3776
	30	1.3156	1.4189	0.6929	0.1423	0.6292	0.5069	0.4173	1.3792	0.6099	4.3987

A distinctive feature of IMOEABS is its symmetric vertical spectral feature coding mechanism, which directly manipulates band indices. In contrast, BIMOEABS represents the binary-coded version of IMOEABS, and IROABS also relies on binary encoding. Binary-coded methods incur additional computational overhead due to decoding and bit-level operations, resulting in longer runtimes. The symmetric vertical spectral feature coding mechanism streamlines genetic operations, accelerates the search process, and improves convergence stability. By avoiding discrete bit transformations, IMOEABS effectively balances computational efficiency and solution quality.

VII. CONCLUSION

This article proposes a novel BS method, IMOEABS, and verifies the quality of the selected bands using different classifiers. First, IMOEABS adopts an entropy-driven operator selection strategy to dynamically choose variation operators based on the state of the population, enhancing adaptability during the optimization process. Second, a nondominated sorting mechanism based on DSCD is introduced to reduce the selection of redundant bands and improve the diversity of solutions. In addition, IE is employed to evaluate the information richness of each band, while Bhattacharyya distance is used to measure interclass separability. IMOEABS selects bands that are both information-rich and highly discriminative based on these two objectives. Furthermore, a symmetric interval transformation vertical spectral feature encoding mechanism is designed to improve search efficiency. Finally, the distribution and quality of the selected subsets obtained by different methods are analyzed. Experimental results on three widely used datasets demonstrate that the band subsets selected by the proposed IMOEABS achieve high classification accuracy under both SVM and KNN classifiers, confirming the effectiveness of the proposed method in the BS problem.

Future work will focus on simultaneously optimizing the classifier and enhancing the BS method to further improve classification performance.

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Heming Jia (Member, IEEE) received the B.S. degree in automation and the M.S. degree in control theory and control engineering from Northeast Forestry University, Harbin, China, in 2006 and 2008, respectively, and the Ph.D. degree in system engineering from Harbin Engineering University, Harbin, in 2012.

He is currently a Professor with the School of Information Engineering, Sanming University, Sanming, China. His research interests include multiobjective optimization, engineering applications, and swarm intelligence optimization algorithms.

Dr. Jia is an Editorial Board Member of *Biomimetics* and *Information*. He was also selected as one of the top 2% scientists in the world.



Changsheng Wen received the B.S. degree from the Department of Information Engineering, Sanming University, Sanming, China, in 2024. He is currently pursuing the M.S. degree with the College of Information and Electrical Engineering, Heilongjiang Bayi Agricultural University, Daqing, China.

His research interests include multiobjective evolutionary algorithms, hyperspectral remote sensing image processing, and engineering applications.

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